Ad hoc semantic reconciliation by cognitive agents: a four-setting synthesis

Programme abstract — Two software systems that must exchange information rarely share a model of the world. The classical answer is to agree a common standard in advance; this programme asks whether two systems that can reason can instead reconcile their divergent models for the occasion — ad hoc, machine-to-machine, with no standard settled beforehand. Across four operational settings in network and service management — configuration, intent, cross-domain provisioning, and observability — it builds a measured harness, scores every reconciliation against a validated gold standard, and varies one master control: where the cognition sits, from two live reasoning agents through one inert to both inert. The through-line the four settings establish is a single thesis. It is cognition that completes a reconciliation. Descriptor methods — matching names, then names plus a gloss — carry it to a ceiling and stop; where both systems can reason, the remainder is closed autonomously, with no agreed model and no human in the loop. A thin, published reference can substitute for cognition, or supply an inert side the facts it lacks — but its reach ends there: it hands over information, never the authority to decide, whose value governs or whether a trade-off is acceptable. And the pragmatic layer — what a reconciled thing is for, whether it matters, who decides — is the frontier the descriptor methods never reach, decisive for meaning and itself bounded by the agent's capability. This document is the synthesis that sits atop the four setting reports: it introduces the concepts once, distils each setting to its essentials, and gathers the findings so they can be read as one result. The full evidence for any setting lives in its own report (1/4–4/4), to which this one points throughout.

Part I — The idea

1. The problem: two models, and no time to standardise

When two software systems must work together but hold different models — of the same domain, or of adjacent domains that must connect — someone has to reconcile those models. The scope a model must cover is not fixed, the useful level of abstraction depends on the task, and capable engineers, working independently, produce different but defensible models of the same network. The classical remedy is a universal standard agreed ahead of time: everyone adopts one vocabulary, and the reconciliation is done once, by committee, before the systems ever meet. That remedy is expensive, slow, and permanently behind the systems it tries to govern.

This programme investigates the alternative that becomes available once the systems on each side can **reason**. Two such systems need not wait for a standard; they can reconcile their models **ad hoc** — between themselves, for the task at hand, machine-to-machine. The question is whether that actually works, how far it reaches, and what — if anything — a small published aid adds to it. The answer is built empirically, in a harness where the reconciling parties are language-model agents and every outcome is scored against a validated gold standard, so that the claims rest on measurement across conditions rather than a single worked demonstration.

2. The lift: from a data model to a portable semantic model

The move the whole programme turns on is the **lift**, and it is worth drawing before anything else (Figure 1).

A system usually holds its content as a **data model** — a schema and its records as they sit in a database or a controller. That content is already a partial semantic picture, but a fixed one, frozen by the format it uses: a field named `och_grade` with values `1`, `2`, `3` is legible to the software and engineers built around its schema, yet says nothing, on its own, to a consumer that was not told what it means — the meaning lives in a specification and in shared convention, outside the data itself. The **lift** is the move from that to a **semantic model**: the same content given grounded, explicit meaning, structured as three parts. First, an **ontology**, which includes its **lexicon** — a schematic layer (the concepts, their kinds and relations, and the preferred labels and synonyms that name them) and a concrete layer (the individual instances that populate those concepts). Second, **pragmatics**: the contextual information a consumer needs — what a thing is for, whose authority governs it, in what context it holds. Third, **provenance**: who asserted each part, by what method, and how firmly. The lift itself is a **cognitive act**, helped along by a few **supports** that are aids to that cognition rather than parts of the model — definitions, worked examples, a canonical example, and an optional link to a shared reference. Who performs the lift is itself a variable of the study: a live system lifts and explains itself; an inert one is lifted by whatever cognition reads it.

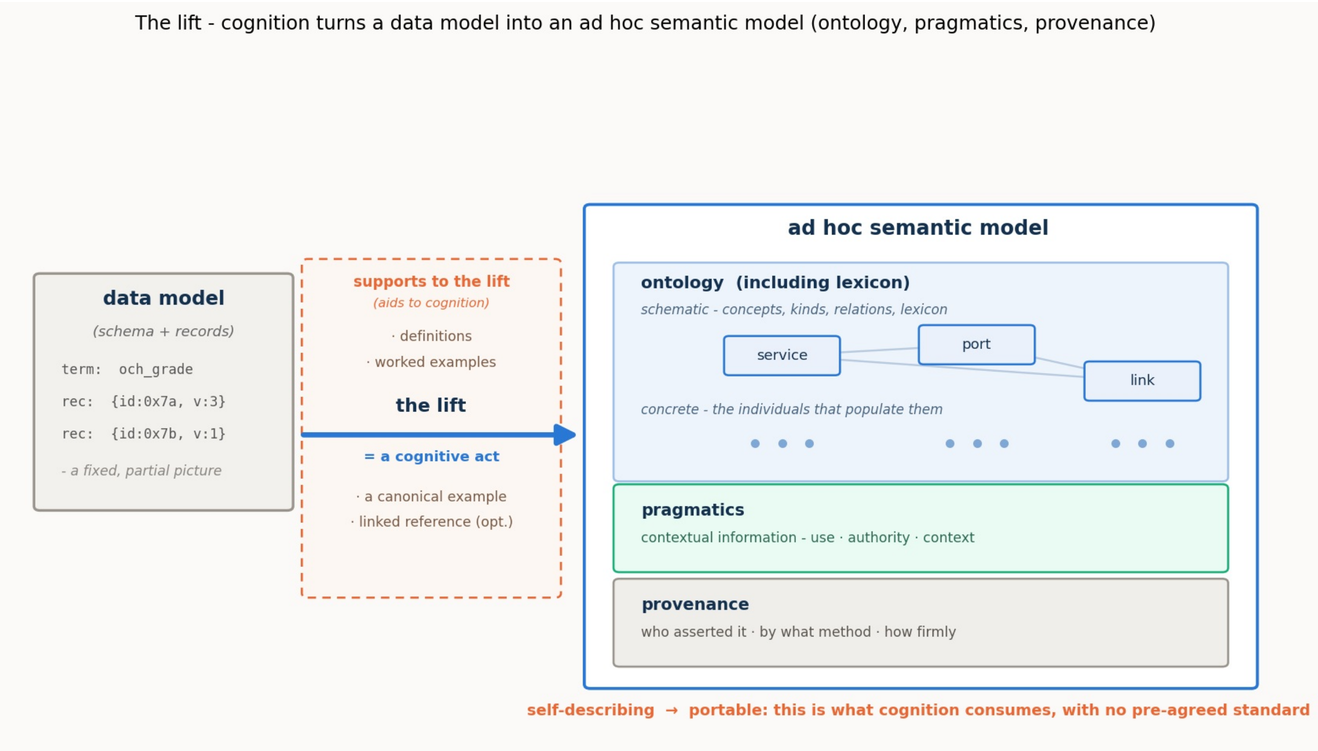

Figure 1. The lift. A data model (schema plus records) is a partial semantic picture, fixed by its format, with much of its meaning left implicit — in specifications, convention, and the engineers who built around it. The lift is a cognitive act, aided by a few supports (definitions, worked examples, a canonical example, an optional linked reference), that turns the data model into an ad hoc semantic model built from three parts: an ontology including its lexicon (schematic concepts and the concrete instances that populate them), pragmatics (use, authority, context), and provenance. Being self-describing, the result is portable: any cognitive consumer can pick it up and understand it, with no pre-agreed standard.

The payload of the picture is the word **portable**. A lifted semantic model is **self-describing**, and being self-describing is exactly what lets *any* cognitive consumer — this agent, another agent, a human — pick it up and understand it without a prior agreement. That portability is not a convenience; it is the precondition for the whole approach. Cognitive consumption of a model — using it, reasoning over it, reconciling it with another — requires meaning made explicit, and the lift is how meaning is made explicit on demand rather than by standardisation. Reconciliation is one instance of cognitive consumption; the settings that follow will also refine an intent against a catalogue, and read an anomaly for its significance, over the same lifted substrate. Get the lift, and the rest of the programme is operations over portable semantic models.

One framing is worth making explicit, because it connects this work to a fast-growing body of practice. The ontology and the individuals that populate it are, together, a **knowledge graph**: the ontology is its schema (the typed concepts and the relations permitted among them) and the instances are its assertions (typed individuals and the edges realised among them). The reconciliation operations that follow divide along the same seam. Aligning the concepts is schema, or ontology, matching; deciding which individual is which is entity resolution over the graph. This is why the community's knowledge-graph work for network operations, and the canonical, AI-facing modelling interfaces now emerging, meet this approach squarely: a lifted semantic model is a knowledge graph a machine can consume, and a thin shared reference is exactly the kind of interoperable anchor such graphs and interfaces can bind to. What the semantic model adds beyond the graph is the pragmatic layer and the provenance base, the facets a bare knowledge graph carries only weakly, and, as the findings will show, exactly the facets a thin reference cannot supply.

3. Reconciliation over lifted models: the operations

Given two lifted models, **reconciliation** aligns them: it works out which concept on one side denotes the same thing as which concept on the other, fixes the shared attributes so they cannot be misread, aligns the individuals, and confirms the result (Figure 2). It is not one act but a small family of **operations**, and naming them once is what lets the findings later be mapped to a precise *where*:

- **Lift** — data model → semantic model, per side (§2).
- **Reference construction** — find or build the thin shared ground the two sides bind through.
- **Schema binding** — align the concepts themselves: the **lexical** operation (names and synonyms) and the **ontological/structural** one (kinds, relations, decomposition).
- **Attribute pinning** — fix the meaning of a shared field, so a *committed payload* rate is not read as a *line* rate, a *bound* not as a measured value.
- **Instance co-reference** — decide which individual on one side is which on the other (entity resolution over keys, attributes, and topology, sometimes only settled by probing the live system).
- **Verification** — confirm a proposed correspondence, by round-trip, by invariant, by a virtual provision-and-read-back, or — where the relation is a refinement — by a satisfaction check.
- **Pragmatic resolution** — settle what the reconciled thing is *for*: whether a degraded offer is acceptable, whose realm governs a shared field, whether an anomaly warrants a **page** (an alert raised to an on-call human operator).
- **Composition (correlation)** — assemble separately reconciled parts into a composite whole: in setting 4, correlating several symptoms into a single incident by following the resource-dependency structure.
- **Lifecycle recurrence** — re-reconcile as the live situation changes over time.

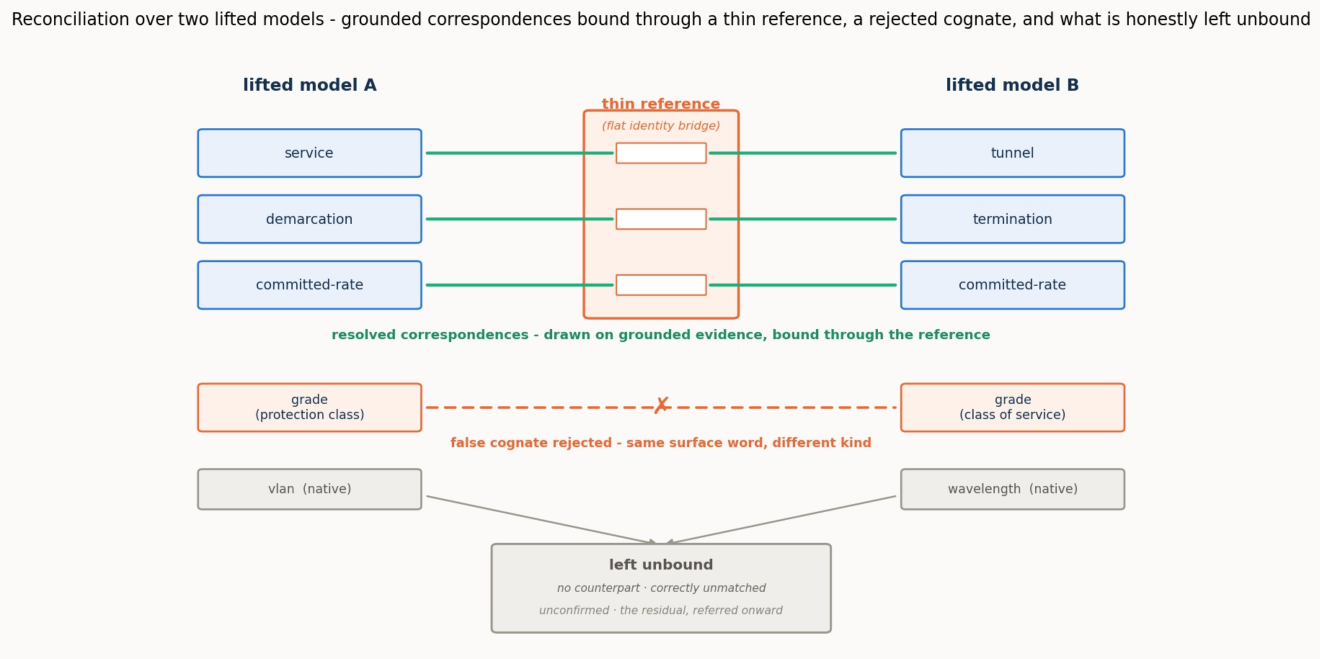

Figure 2. Reconciliation over two lifted models. Correspondences are drawn on grounded evidence and bound through a thin reference that acts as a flat identity bridge; a look-alike that shares only a surface word is rejected as a false cognate on its kind, attachment, and instances; a concept with genuinely no counterpart in the other model is correctly returned as unmatched — a resolved outcome, not a gap; and a correspondence the evidence cannot yet *confirm is left in the residual, referred onward. The reference carries no structure of its own; it is parasitic on the two grounded models it connects.*

Two features of Figure 2 recur in every setting. The first is the **false cognate**: two concepts that share a surface word but denote different things — an optical *signal-grade* and a commercial *service-grade*, a transport *grade* and an IP *grade*, a legacy *alarm* and an NMOP *anomaly*. Reconciliation must draw the true correspondences *and* refuse the look-alikes, and it refuses them on grounded, non-lexical evidence — kind, attachment, instances — which a purely lexical matcher, keying on the shared label, cannot do. The second is the **residual**: the correspondences a pass does not close. A residual is not an error; it is what a reconciliation honestly *refers onward* rather than guessing — to further machine cognition where the agents are live to continue, or to a person where they are not. How large the residual is, and what drives it, is one of the programme's central measurements.

4. The cognition spectrum, and the residual as a shortfall

The master control across all four settings is **where the cognition sits** — the **cognition spectrum**. It runs from both sides live and interrogable, through one side inert, to both inert:

- **both-cognitive** — each side is a live reasoner, the authority on its own model, able to explain itself and answer questions;
- **one-inert** — one side is a mute snapshot exposing only its structure and instances; the live side must reconstruct its meaning;
- **both-inert** — neither side can explain itself; a third party reconstructs both from structure and data, and can only *propose* candidates for external adjudication.

A structural fact about this spectrum organises every result that follows. In the fully-cognitive case, resolution of any reconciliation question is total *in principle*, for two reasons that are themselves functions of live cognition on each side. First, the agents can exchange **unbounded further information** — each is the live authority on its own model, so whatever is ambiguous the other can ask about and get an authoritative answer. Second, because reconciliation operates on the models rather than the live network, the agents can **run decisive experiments in virtual space** — provision a candidate through a proposed correspondence, operate it, read it back, and check the invariants. Any question is therefore confirmed, refuted, or authoritatively decided, and nothing need be left unresolved. Both mechanisms are functions of live cognition, so as cognition recedes they fall away: with one side inert the live agent can probe but not interrogate or co-design an experiment; with both inert there is no one to ask and no joint experiment to run. The **residual** a reconciliation must leave and refer onward is, in exactly this sense, the **shortfall from full cognition's reach** — and it grows as that reach recedes. It is not a fixed floor the fully-cognitive case merely reaches; between two fully-cognitive agents there is, in principle, none.

This is worth stating as a finding in its own right. Across the four settings we found no semantic gap that a sufficiently capable, sufficiently reaching cognition could not close. What remains once two fully-cognitive agents have exchanged everything they can and run every decisive virtual experiment is never an *unbridgeable* correspondence: it is a concept with genuinely no counterpart (correctly returned as unmatched), a fact not yet realised in the running network (an absence in the world, not in meaning), or a question of *authority* rather than of fact (§5). None of these is a gap cognition is stuck on — and the first two a human reasoner would leave exactly where a machine does. What varies from one agent to the next is therefore not the *kind* of cognition but its power and reach, which is precisely what the capability gradient (§13.2) and the reach study (§13.5) measure.

This is the frame inside which every finding sits, and it carries the programme's central claim. Lexical and descriptor methods carry a reconciliation to roughly ninety percent — names, then names plus a gloss — and there such methods have historically stopped, the remainder left to an agreed standard or to human judgement. What the spectrum shows is that the remainder need not wait for either: where both systems can reason, the reconciliation completes **autonomously**, and only as cognition recedes does closing the gap fall back to a reference or a person. **It is cognition that completes a reconciliation**, and the cognition spectrum is the measure of how far the automation reaches before it must hand off.

5. The thin reference: substitute, supply, and the limit at authority

Against that frame, a thin, published **shared reference** is switched on and off at each point on the spectrum — as much a probe of what cognition was doing as an object of study in its own right. The reference is deliberately small: a flat set of entries, each an identity anchor plus a few descriptive fields (a label and synonyms, a shallow class, a one-line definition, a canonical example). It has no ontology of its own — giving it one would turn it back into the universal standard the approach exists to avoid — and its power is coordination, not content: two systems that bind to the same entry thereby denote the same thing. It is **parasitic** on the two grounded models it connects, which is precisely why it can be so thin.

What such a reference does, across the settings, resolves into three distinct roles, and keeping them apart is essential:

- It can **partly substitute for cognition**. For a capable agent on a lexical/structural task, the reference does the reasoning's work — collapsing the agent's hidden deliberation by one to two orders of magnitude and yielding a perfect, verified reconciliation.
- It can **supply missing information**. Where a side is inert, the reference can hand the reading agent facts it can no longer obtain by interrogation — a committed guarantee, a unit, a categorical definition — restoring the verification the missing probe would otherwise have done.
- It **cannot supply authority**. Where what is missing is not a fact but a *judgement* — whether a degraded offer is acceptable, whose realm governs a contested field — no reference substitutes. A reference can tell you what a service guarantees; it cannot tell you whether this customer will accept it. Where the gap is information, a thin reference closes it; where the gap is authority, only cognition — live, or pre-placed as a policy — closes it.

The reference's value is also **capability-gated** and distributed unevenly along the spectrum. Its effort benefit peaks with strong, live cognition — a weak agent may not be able to exploit it, and an inert side can turn the extra material into a burden. Its error-prevention benefit runs the other way, concentrating where cognition, and therefore verification, is weakest. One anchor, opposite roles, and — a finding the settings sharpen — a definite **internal structure**: a reference's fields are not of equal value, and the cheapest useful one is not the cheapest possible one.

6. The instrument

Every setting is measured in the same harness. A **case** is two lifted semantic models plus a gold-standard reconciliation *derived from the models by a script and validated* before any run, so the gold cannot drift from the models it scores. A **reasoning stack** stands in for the reconciling cognition — in the fully-cognitive case, an exchange between two live agents; with a side inert, the live agent reconstructing the mute one; with both inert, a third party — and the harness scores the stack's output against the gold and records both quality and cognitive effort. Quality is measured by **precision** (of what it proposes, the fraction correct), the **resolved fraction** (of the true correspondences, the fraction it actually commits, the rest referred to the residual), **surviving false cognates** (planted traps taken), and the **residual** itself; effort, for a language-model stack, by **reasoning tokens** (hidden deliberation), total tokens, and latency. Throughout, a resolved fraction below one is not by itself a failure — it is reach deliberately traded for honesty, the unconfirmed correspondences referred onward rather than guessed.

The reconciling agents are run over one **model ladder** for the whole programme — three models spanning a capability range, `gpt-5.6-sol` (**strong**), `gpt-5-mini` (**mid**), and `gpt-5-nano` (**weak**), referred to as **sol**, **mini**, and **nano** — chosen so the ladder isolates model strength rather than provider or architecture. The programme's hypotheses are the ones the first setting states and the others inherit: that cognitive agents can reconcile divergent models ad hoc (the concept holds); that the placement of cognition governs what a reconciliation achieves, costs, and can verify; that a thin reference partly substitutes for cognition, capability-dependently, and prevents errors where cognition is weakest; and that reconciliation work scales linearly with a shared reference against quadratically without one. What each setting adds is a new operation brought under test — verification and instances, then pragmatics as a movable policy, then the standard-free bind, then pragmatics as significance, whether an observation matters at all — against the same frame and the same instrument.

Part II — The four settings

Each setting takes the same frame to a new operation. The sections below are distilled to a common rhythm — what is new against the first setting, the case in one paragraph, the operations put under test, what was proven, and what it means — and each points to its own report for the full evidence.

7. Setting 1 — Configuration: two standard models of one network

Full report: 1/4 · Configuration.

What it establishes. This is the founding setting: it puts the whole idea to its first empirical test and fixes the frame the others inherit. Two independently authored **public standard** models of one optical transport network — ONF **TAPI** on one side, IETF **TEAS/ACTN** on the other — describe the same nodes, links, and services, each in its own vocabulary. A TAPI *connectivity-service* is a TEAS *tunnel*; a TAPI *link-termination-point* looks like a TEAS *tunnel-termination-point* but is a different thing — a link end, not a trail head. Because both models are public standards the agents recognise, cognition can lean on that recognition to bridge them, which makes this the cleanest place to isolate the mechanism.

Operations under test. The reconciliation as an **equivalence** (this term *is* that term): the **lexical** and **ontological/structural** binding, **verification** run as its own step, and **instance co-reference** — with the **pragmatic** operation left untouched here, deferred to the later settings. The thin **reference** is the lexical one, switched on and off across the spectrum.

What was proven. The concept holds, measured across conditions rather than shown once: cognitive agents reconcile the two models correctly and ad hoc. Within that, the reference **substitutes for cognition** — for the strong agent it collapses hidden deliberation by one to two orders of magnitude and yields a perfect, verified reconciliation (precision and resolved fraction of one) across the whole spectrum. The benefit is **capability-dependent** and not monotonic: for a weak agent the reference can *add* effort when a side is inert, and the naive "reference helps more as cognition recedes" hypothesis is refuted *for effort* even as it holds *for correctness* — precision separates the models on the hard case (strong 0.94→1.00, mid 0.88→0.96, weak 0.91→0.97 with the reference) and the reference prevents the errors a weak agent otherwise commits. **Verification** runs as its own step, keeping only the correspondences it can confirm and referring the rest onward. It rejects the false cognates that slipped into the proposal, so precision climbs toward one across the spectrum. But confirming a correspondence draws on live cognition, which the spectrum removes. Without the reference, as a side goes inert, the verifier can still reject a wrong pairing yet can no longer confirm every correct one; those unconfirmable-but-correct pairings are deferred to the residual rather than asserted, so the resolved fraction falls. The reference supplies the missing confirmation, holding the resolved fraction at one. Reconciliation work scales **linearly** with a shared reference against **quadratically** without one (verified by construction to N = 12). And two folded sub-studies carry the point past the schema terms: at the **instance** level the resolvability of same-looking individuals is budget-limited at full cognition (driven to a full resolve with more probing) but becomes **structural** once a side goes inert (no budget helps, because the inert side cannot be interrogated); and a **pre-lift baseline** confirms the lift itself is the lever — matching on the bare lexical surface leaves a quarter to a third of the correspondences unfound (resolved fraction 0.66–0.76), and restoring the lifted content recovers them (0.92–0.97).

What it means. The founding claim is in hand: cognition completes the reconciliation, a reference partly substitutes for it, saving cognitive effort and preventing errors, and the spectrum governs both. Everything after this builds on that frame; setting 1 does not touch pragmatics at all — that is the thread the remaining three settings pick up.

8. Setting 2 — Intent: refinement, negotiation, and a service that renegotiates itself

Full report: 2/4 · Intent.

What is new. The first setting reconciled by **equivalence**; this one reconciles a declarative **intent** against a concrete **realisation** by **refinement**. "Latency below 5 ms" is not equal to any service on offer — it is a *bound* that a service either clears or does not, and many services may clear it. Once the relation is refinement, everything the first setting left untouched comes into play at once.

The case. A customer's agent **O** holds an intent — a New York–Frankfurt connection, ≥ 8 Gbit/s, ≤ 5 ms, four-nines, protection not required — every clause a bound. An operator's agent **N** holds a catalogue of concrete optical services, each with a real bandwidth, latency, availability, protection scheme, and cost. Neither speaks the other's language; they must work out which realisation *satisfies* the wish.

Operations under test. Verification becomes a **satisfaction** check (you cannot round-trip a lossy refinement, so you test the chosen realisation against every bound); a genuine two-sided **negotiation** appears when no service meets every bound (N computes a best-achievable offer, O must decide accept or reject); the **pragmatic** operation enters for the first time, carried in a small, portable **movable policy** the customer holds — a priority ordering, hard bounds, an affordability floor, a flow-class rule; and the whole exchange **recurs across the service's life**.

What was proven. The central insight, now measured on a real negotiation: with both agents live, the reconciliation — negotiation included — completes autonomously (decision accuracy 1.0 for sol and mini), with no human in the loop. The spectrum gains a new rung that turns out to be the sharpest result: remove the customer's live judgement, and a **pre-placed movable policy** still closes the decisions it was authorised for (1.0), while a **mute** customer with no policy cannot — accuracy falls to zero and the agents correctly **refer all three** decisions onward rather than guess. The pre-placed policy is a concrete mechanism for **pushing the hand-off boundary**: the customer's cognition, placed in a portable artefact ahead of time, closes autonomously what a mute description must refer. The second finding draws the reference's limit: where a side is inert, a published reference *supplies information* (the strong agent, blind at both-inert, refuses to affirm satisfaction and scores 0.29; publish the invariant guarantee floor and it has an anchor to check against, rising to 0.71) — but it *cannot supply authority*: at both-inert no reference moves the negotiation, because what is missing is the customer's judgement itself. The capability gradient is steep (to reach its decisions the strong agent spent ~150 reasoning tokens, the mid ~860, the weak ~5,400 — thirty-five times the strong agent's effort, for lower accuracy), and a four-hop **lifecycle** — bought, self-healed, referred, restored — is walked correctly end to end by the strong model (hop accuracy sol 1.0, mini 0.88, nano 0.62).

What it means. Cognition completes a negotiation, and the pragmatic operation is now a measured axis rather than a fixed backdrop. A thin published reference can stand in for the information an inert side would otherwise supply, but not for the authority to decide. A pre-placed policy is shown able to carry a party's authority to where a person or live cognitive system would otherwise have to stand.

9. Setting 3 — Cross-domain: reconciling with no public standard

Full report: 3/4 · Cross-domain.

What is new. The deliberate complement to the first setting. There, two *different* models faced each other — but both were **public standards** the agents already knew, so cognition could lean on recognition. Here that crutch is gone: both models are **home-grown and private**, recognisable to no one in advance. The operations are the first setting's — lift, bind, a thin reference — but with the standard removed, the *execution* diverges, and that divergence is the subject.

The case (Figure 3). Meridian, a home-grown transport OSS, thinks in circuits, bearers, wavelengths, protection grades. **Cascade**, a home-grown IP/VPN controller, thinks in services, attachments, VLANs, classes of service. Their worlds touch at exactly one seam: a Cascade service must ride a Meridian circuit as its underlay. One order across that seam needs five bindings — circuit↔underlay and hand-off↔attachment (the same object under two names), and three requirements pinned (a *committed* rate not a line rate, a latency *bound*, protection against a *path* failure) — and one look-alike refused: both models carry a **grade**, but Meridian's is a transport protection class and Cascade's an IP class of service. Same word, unrelated meanings, and no standard to consult.

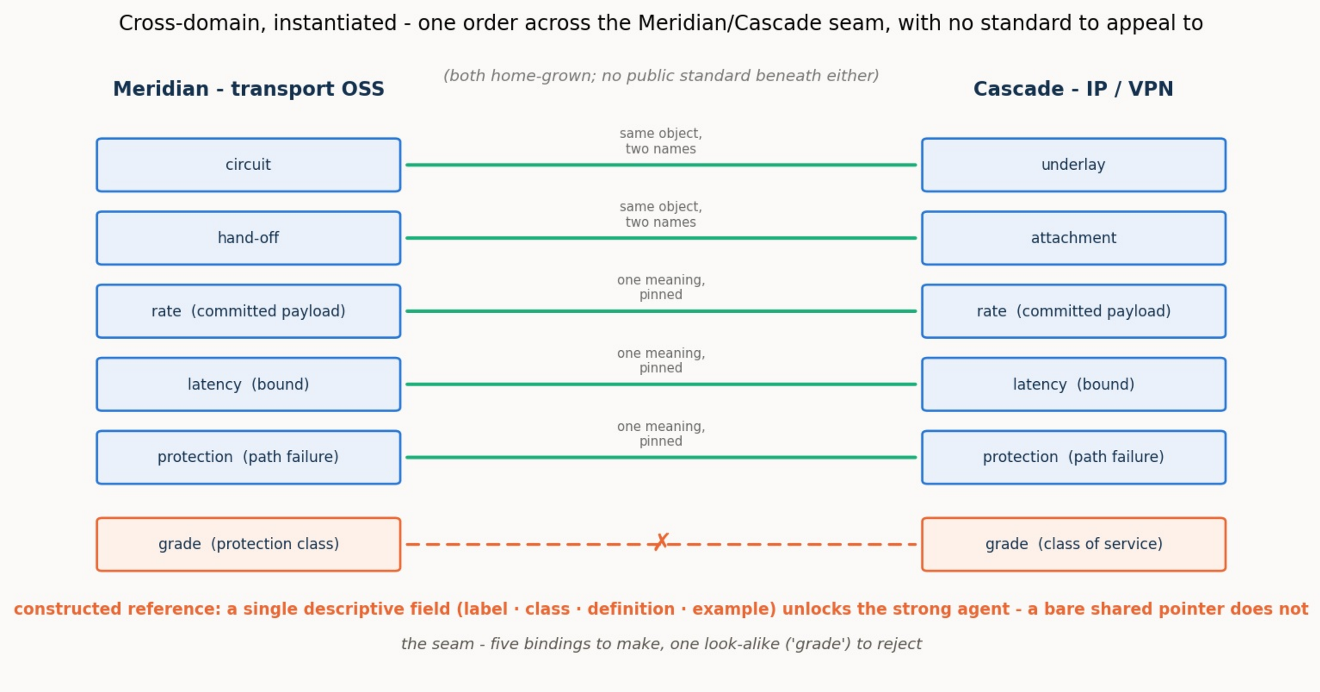

Figure 3. One order across the Meridian/Cascade seam. Five bindings to make — two renamings and three pins — and one false cognate, "grade", to reject, across two private vocabularies with nothing public beneath them. A constructed reference supplies the shared ground; a single descriptive field in it is enough to unlock a capable agent.

Operations under test. A single, early **schema-binding** pass — deliberately isolated from the rest of the process — measured across the spectrum with and without the reference the two agents would themselves **construct**; plus a **reference-field ablation** and the **pragmatic** operation of authority attribution (whose realm owns each shared field). The instance operation is bracketed, on the argued grounds that it reproduces the first setting's result.

What was proven. The central result is a pair of **mirror-image failures** — the strong and weak agents breaking in opposite, symmetric ways. With the constructed reference absent, the **strong agent under-commits and the weak agent mis-commits**. The strong agent will not guess across two foreign vocabularies — it binds only the names that already coincide (resolved fraction ~0.5 at both-cognitive, ~0.4 once a side is inert) but at perfect precision and with no false cognate taken: this is **omission**, deferral, not error. The weak agent does the opposite — it binds freely (resolved fraction 0.8–1.0) at a precision that never clears 0.83 and falls to 0.57 inert, taking the cross-domain "grade" trap: **commission**, confident error. A single thin reference remedies both: constructed, it lifts the strong agent to a full close and lifts the weak agent's precision toward one. Stripping the reference's fields one at a time shows what it must carry: any *single* descriptive field — a shared label, a class, a definition, or an example — is enough to unlock the strong agent's commitment (each drives it to a near-perfect close), while a **bare shared identifier with no description is worse than nothing** (precision 0.50, below the no-reference floor), because the agent binds by an opaque token and binds wrongly. The reference does not work through the shared *pointer*; it works through the shared *description*, and even the thinnest description suffices. On the **pragmatic** axis, the same boundary the intent setting found appears again: the reference reaches *meaning* but not *authority* — it can pin that a rate is a committed payload, but not whose realm governs it — and a characteristic "transport owns everything it carries" bias marks the weaker models.

What it means. With no public standard beneath two models, cognition still closes — but only once it has built the shared ground, and **building that ground is the work**. The strong agent's low reference-absent numbers are not a limit of cognition; they measure the worth of the one step the study held back — constructing the shared reference — by running the binding pass without it. Run end to end — the agent constructing the shared reference itself from the two models and then binding through it, with none pre-given — the protocol confirms the reading: the strong agent lifts from a no-reference resolved fraction of 0.40 to a constructed-reference **0.93**, approaching the reference-given 1.00, at perfect precision and with no false cognate, so constructing the ground is the work and a capable agent does it. Even a very thin ground suffices for a capable agent, provided it carries meaning and not merely a pointer.

10. Setting 4 — Observability: an alarm is not an anomaly

Full report: 4/4 · Observability.

What is new. The setting where the pragmatic operation moves from the wings to the centre. The descriptor level — what a thing *is* — is settled by a standard, and the entire operational question is one of **significance**: whether an observed deviation matters, and how observations compose. It also carries the programme's deepest false cognate, an **ontological** one.

The case (Figure 4). Agent F, a legacy fault manager, emits an **alarm**: one object bundling an event, an undesirable state, a fixed severity, and a static probable-cause. **Agent G**, an IETF **NMOP** agent, speaks the RFC 9940 ladder, which separates what legacy conflates — event, anomaly, symptom, fault, alarm, problem, cause, incident — and annotates each anomaly with anomaly-semantics metadata (a concern score, a confidence score, a plane, a pattern, a lifecycle stage, a season). The worked example is one live anomaly: a pre-FEC bit-error-rate reading on a wavelength begins to rise. In the legacy world it fires an alarm and a page; in the NMOP world nothing is decided yet — whether it warrants a page depends on its concern and confidence and its context (a maintenance window makes the same deviation expected), and whether it is one incident or many depends on correlating it, across layers, with the symptoms it causes.

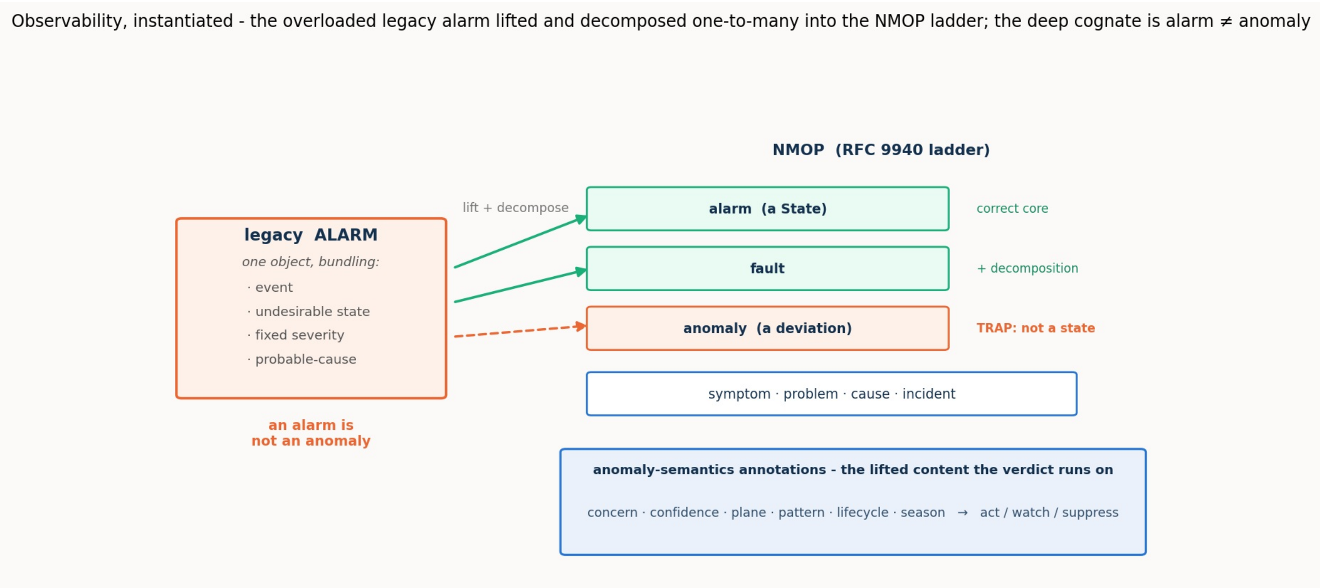

Figure 4. The overloaded legacy alarm is lifted and decomposed one-to-many into the NMOP ladder: its correct core is the NMOP alarm-State (and the fault it implies), while the trap is the anomaly — an alarm is a State, an anomaly is a deviation, and conflating them is a category error, not a mislabel. The anomaly-semantics annotations are the lifted content the significance verdict runs on.

Operations under test. Two acts. **Act 1** is a **schema binding** carrying the ontological false cognate (alarm↔anomaly, which no structural cue separates) and a **one-to-many decomposition** (the legacy alarm maps to both the NMOP alarm-State and the fault). **Act 2** is pure **pragmatic resolution and composition**, each run with the anomaly-semantics **ON** and **OFF**: a **verdict** task (act / watch / suppress for each anomaly under a context) and a **correlation** task (group cross-layer symptoms into incidents and name each cause).

What was proven. Three results complete the arc. First, the **ontological cognate is a clean three-rung capability gradient**: the strong agent never conflates alarm with anomaly (with or without the reference); the mid agent conflates them once a side is inert, and the RFC 9940-anchored reference **rescues it completely** (the cognate vanishes, precision returns to 1.0); the weak agent conflates them with or without the reference — beyond rescue. The lexicon pins the ontology for the middle of the ladder, not the bottom. (The honest hard edge of Act 1 is the decomposition: even the strong agent tends to map the alarm to the alarm-State but miss the fault constituent, so the resolved fraction sits near 0.75.) Second, in the verdict task the **pragmatics carry the operative meaning** — with the semantics ON the strong and mid agents suppress correctly during maintenance and the false-page storm disappears (verdict accuracy 1.0 and 0.83, essentially no false pages); with them OFF the legacy pipeline pages nearly everything (accuracy 0.17 and 0.08, roughly four and three-and-a-half false pages) — **but the payoff is capability-gated**: handed the identical annotations, the weak agent barely moves (ON 0.53 vs OFF 0.50). Third, **correlation behaves differently**: given the resource-dependency map, *every* model — the weak one included — folds an optical degradation and the IP loss it causes into one correctly rooted incident (perfect partition ON), and without that map every model fails. The difference is the lesson: correlation's pragmatic is a **structural input** and even a weak agent applies it; the verdict's pragmatic demands **judgement**, and there capability decides.

What it means. Meaning (what an anomaly *is*, pinned by the reference) and significance (whether it warrants a page and how it composes, carried by the pragmatics) are **separable, both necessary, and each gated in its own way**. The pragmatic layer the descriptor methods never reach is real and decisive — and itself bounded by the agent's capability. The through-line reaches its end: cognition completes the reconciliation; a thin reference supplies the information it needs and stops at what it does not; and the pragmatic layer is the frontier, decisive for meaning and gated by capability.

Part III — Synthesis

11. What works, and how far

Read as one result, our detailed exploration of the four settings says first the plain thing: **ad hoc reconciliation works** — largely as laid out, barring a few surprises. Two cognitive agents reconcile independently authored, divergent models correctly and with no standard agreed in advance — completing an equivalence between two standard models (setting 1), refining and negotiating an intent against a catalogue (setting 2), binding across a private domain boundary once they have built the shared ground (setting 3), and reading an observability world for what its signals mean (setting 4) — measured against a validated gold across the cognition spectrum, not shown once by hand.

The same exploration says, more sharply, **where the reconciliation needs no help**. At the fully-cognitive end of the spectrum it completes **autonomously in every setting** — no standard agreed in advance, no human in the loop — including the two operations that look least automatable: a two-sided negotiation (setting 2, decision accuracy 1.0 with both sides live) and deciding, for each anomaly, whether to act on it, watch it, or suppress it — the significance verdict, judging whether an observed deviation actually matters (setting 4, accuracy 1.0 with the pragmatics on). The fully-cognitive end is, across all four settings, the automatable end. That is the headline, and everything else in this synthesis is a qualification of it: how the automation degrades as cognition recedes, what a thin reference buys back, and where an agent too weak for the task cuts the whole thing off.

12. The findings, as six theses — and mapped to where they act

The results gather into six theses. Each is stated once here; §13 then reads the evidence behind them along four cross-cutting axes, drawing the setting reports' own figures and numbers; and §14 lays the whole out in two tables — so that when a finding says cognition (or a reference, or a pragmatic) matters "here," the *here* is a named stage of the process, not a vague gesture.

Thesis 1 — It is cognition that completes a reconciliation: descriptor methods carry it most of the way, then reach a ceiling and stop. Lexical and descriptor matching carry a reconciliation to roughly ninety percent (setting 1's baseline: lexical surface 0.66–0.76, the lift recovering it to the mid-nineties); the remainder, historically left to a standard or a person, is closed by live cognition instead. This is the programme's spine, and it holds at every operation the settings put under test. It holds against a strong classical matcher, not only the plain label baseline: a matcher using labels, synonyms, definitions, and structure with a 1:1 alignment reaches precision one and refuses the false cognates on the standard cases, but it cannot close — its resolved fraction stalls at 0.56 to 0.75 — and on the standard-free case it fails like the weak baseline, taking the *grade* cognate, because with two private vocabularies there is no lexical or structural signal to lean on. The ceiling is real even for a good descriptor method; cognition is what passes it.

Thesis 2 — The placement of cognition is the master variable: the further it recedes, the more the reconciliation leaves unresolved. What a reconciliation can achieve, cost, and verify is governed by where the cognition sits. Between two live agents, resolution is complete in principle — unbounded interrogation and decisive virtual experiment — so the residual is, in principle, none; as a side goes inert those mechanisms fall away and a residual appears, precisely the shortfall from full cognition's reach. The same story holds at the schema level (setting 1), the instance level (setting 1's budget-limited-becomes-structural curve), the negotiation (setting 2), and the standard-free bind (setting 3). And it governs one thing that cuts across all four settings: whether a reconciliation can be **verified to completion**. This is the crux of the spectrum. With full cognition on both sides, the very mechanisms that close the residual — mutual interrogation and decisive virtual experiment — also confirm the close, so the agents verify their own result rather than assert it; as a side goes inert those mechanisms fall away, verification degrades to a check by satisfaction, and the assurance that the reconciliation is correct weakens with it. Full cognition is therefore not merely more accurate: it is what makes an ad hoc reconciliation **self-verifying**, and that — not accuracy alone — is what makes it safe to automate with no standard and no person in the loop.

Thesis 3 — A thin reference partly substitutes for cognition and supplies information, but never provides the authority to decide. For a capable agent on a lexical/structural task it substitutes for the reasoning (setting 1, effort down one-to-two orders, a perfect verified close); where a side is inert it supplies the facts interrogation no longer can (setting 2, sol 0.29→0.71 with the invariant floor; setting 4, the RFC 9940 reference rescuing the mid agent's ontology); but where the missing ingredient is judgement rather than fact, no reference moves it (setting 2's authority gap, setting 3's authority attribution). Information has a published stand-in; authority does not.

Thesis 4 — Strong and weak agents fail in opposite directions: strong agents omit, weak agents commit. Denied the ground it needs, a strong agent **defers** — it leaves the unresolved in the residual at perfect precision (setting 3's under-commitment; setting 2's strong agent refusing to affirm what it cannot verify). A weak agent **commits** — it binds freely and wrongly, takes the false cognate, and is no less confident on a wrong answer than a right one (setting 3's mis-commitment; setting 1's traps surviving for the weak agent). The strong agent's error is a larger residual; the weak agent's is lower precision. This is why a reference does two different things at the two ends of the spectrum. Its **correctness value** — pulling right the bindings that would otherwise go wrong — concentrates where cognition is weakest: the weak agent is the one that commits false cognates, and the reference is what disciplines it out of them. Its **effort value** — the reasoning it saves — peaks where cognition is strongest: a capable agent that would have reasoned its own way to the answer can instead lean on the reference and spend one-to-two orders less to get there. The strong agent needed the economy, not the correction; the weak agent needed the correction, and could not always use it. One thin artifact, a different benefit at each end of the spectrum.

Thesis 5 — The pragmatic layer — what a thing means in context and whether it matters — is the frontier the descriptor methods never reach: decisive and itself gated by cognitive power. What a reconciled thing is *for* — whether a degraded offer is acceptable (setting 2), whose realm owns a field (setting 3), whether an anomaly warrants a page (setting 4) — is where the operative meaning lives, and it is beyond the reach of names, glosses, and references. It is carried by cognition (or by cognition pre-placed as a policy), and its payoff is realised only by an agent strong enough to carry it — a matter of the agent's power, not of where cognition is placed: handed identical annotations, the weak agent still cannot produce the verdict (setting 4). The one exception proves the rule — where a pragmatic is delivered as a **structural input** rather than a judgement (setting 4's correlation dependency map), even a weak agent applies it.

Thesis 6 — With a shared reference, reconciliation scales; and its potential fields are not of equal value. Work grows linearly in the number of systems with a shared reference against quadratically without one (setting 1, to N = 12), so the reference's advantage compounds with scale independent of any per-reconciliation effect. And not every thin reference is equal: a single descriptive field unlocks a capable agent, but a bare identifier with no description is worse than nothing (setting 3), and a shallow class tag can actively mislead the weak agent it was meant to help (setting 1's field-by-field result). A reference is a safety rail carrying *meaning*, not a pointer and not a payload.

13. The findings in depth: four readings across the settings

The six theses state *what* was found. This section shows the evidence behind them, drawn from the setting reports' own figures and numbers, and reads that evidence along four cross-cutting axes: the **cognitive load** an operation costs, the **model power** the gradient demands, what the **reference** buys, and what changes with **position on the cognition spectrum**. Each axis cuts across all four settings, and together they are the detail the one-line theses compress.

13.1 Cognitive load — what each operation costs, and who pays

The most consistent number in the programme is what a weak agent costs. To reach the *same* decisions the weak agent spends one to two orders of magnitude more reasoning than the strong one, in every setting that measures effort (Figure 5): about **36×** in the intent negotiation (sol ~150 tokens, mini ~860, nano ~5,400) and about **20×** in the observability verdict (sol ~60, mini ~520, nano ~1,200). Capability buys economy as much as correctness — a weak agent does not merely make more mistakes, it burns far more cognition making them.

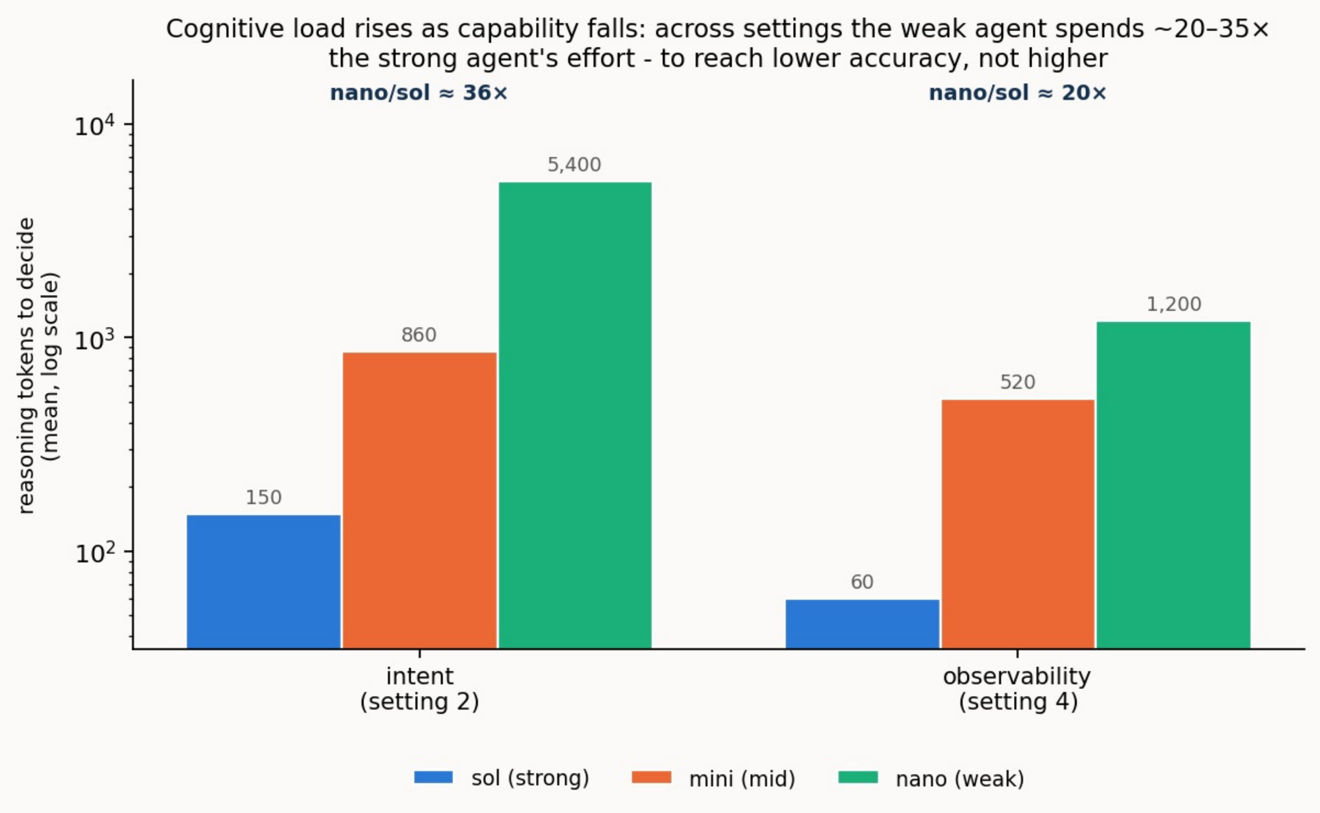

Figure 5. Reasoning tokens to reach a decision, per model, in the two settings that measure effort directly (log scale). The gradient is steep and consistent: the weak agent spends ~20–35× the strong agent's effort — and, as §11 noted, to reach lower accuracy, not higher.

The load also depends on the **operation** and on **reference support**. Where the operation is a lookup and cognition is fully present it is cheap; where it demands search — interrogating a live oracle to separate look-alike individuals (setting 1's instance study) — or judgement — weighing a degraded offer, or an anomaly's concern against its context — it is dear. And for a strong agent a thin reference is an effort *substitute*: given the anchor, sol's hidden deliberation collapses by one to two orders of magnitude (Figure 6). But that effort saving is capability- and placement-dependent, and it can invert: for the weak agent with a side inert, the reference *adds* effort — nano spends roughly 5,500 tokens **more** with the reference than without, straining to reconcile an inert side's meaning against the extra material. The reference's effort benefit peaks with strong, live cognition and turns to a burden at the weak, inert corner.

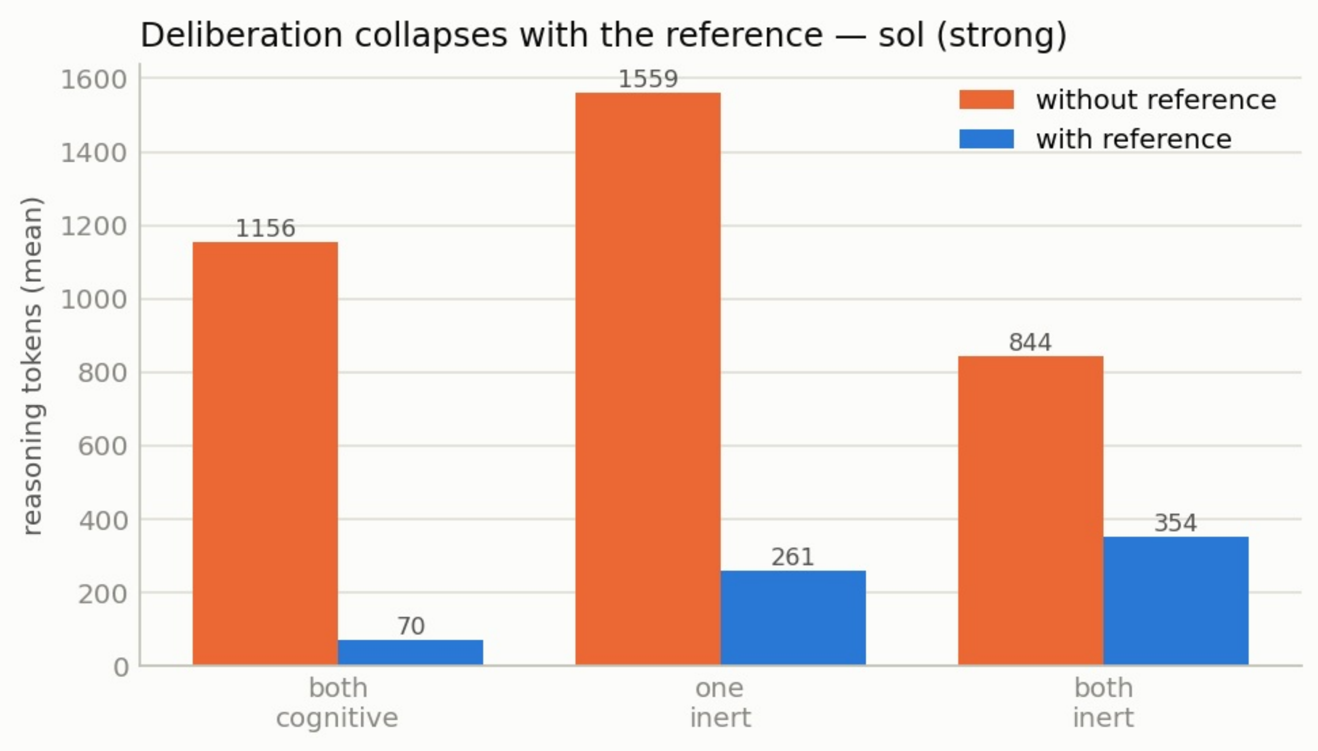

Figure 6 (setting 1). The strong agent's reasoning tokens with and without the reference, across the spectrum. Given the anchor, deliberation collapses by one to two orders of magnitude — the reference doing the reasoning's work.

13.2 Model power — the shape of the capability gradient

Capability does not turn a single dial; it changes the *kind* of failure. The cleanest place to watch this is **setting 3, the cross-domain bind** (Figure 7): Meridian, a transport OSS, and Cascade, an IP/VPN controller, are two independently authored private models with **no public standard between them**, so no ready-made reference exists — the agents must construct the shared ground themselves. Run that bind with the reference withheld and capability alone decides the outcome. Denied the shared ground it needs, the strong agent **omits** and the weak agent **commits** — a mirror. Facing the two foreign vocabularies with no constructed reference, sol binds only the names that already coincide and leaves the rest in the residual: a low resolved fraction, but perfect precision and no trap taken. nano does the opposite, binding freely and wrongly at a precision that never clears 0.83. The strong agent's shortfall is a larger residual (deferral); the weak agent's is lower precision (error); and a single thin reference remedies both, pulling each toward the top-right corner.

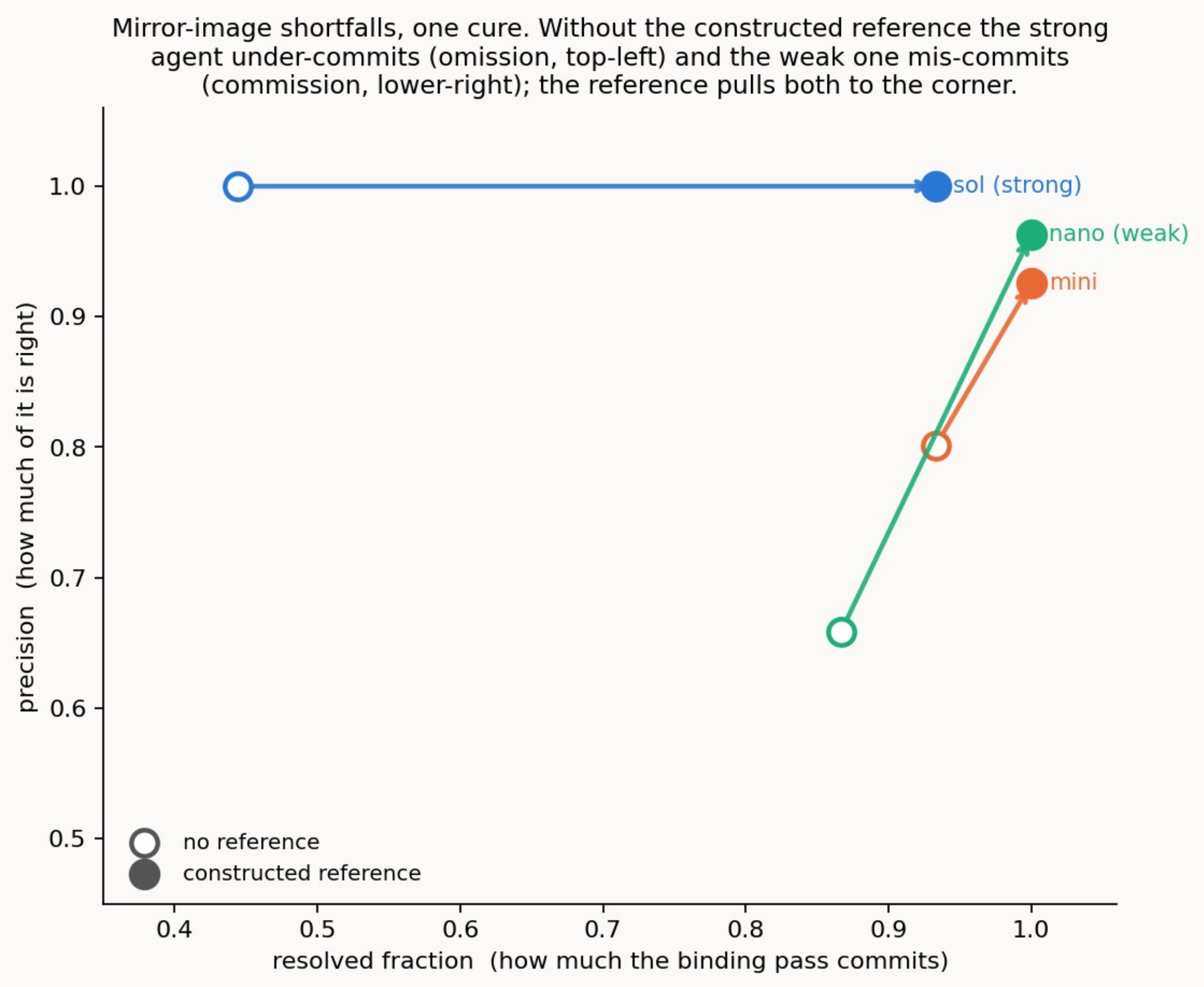

Figure 7 (setting 3). Precision against resolved fraction, reference off (hollow) to on (filled). Without the constructed reference sol sits top-left — commits little, all of it right — and nano lower-right — commits much of it wrongly; the reference pulls both to the corner.

This mirror is not a binary of strong against weak but a smooth gradient. Swept across a six-model ladder on the schema cases (Figure 8), precision falls (0.98 to 0.83) and surviving false cognates rise (from none to most) as capability drops, while resolved fraction rises the other way (0.78 to 0.94): the strong end's omission shades continuously into the weak end's commission, the strongest tier all deferring at near-perfect precision and the failure emerging through the smaller models. What moves along the ladder is not whether the agent can match — it can at every rung — but the quality of its judgement: when to commit, when to defer, and whether it refuses the trap.

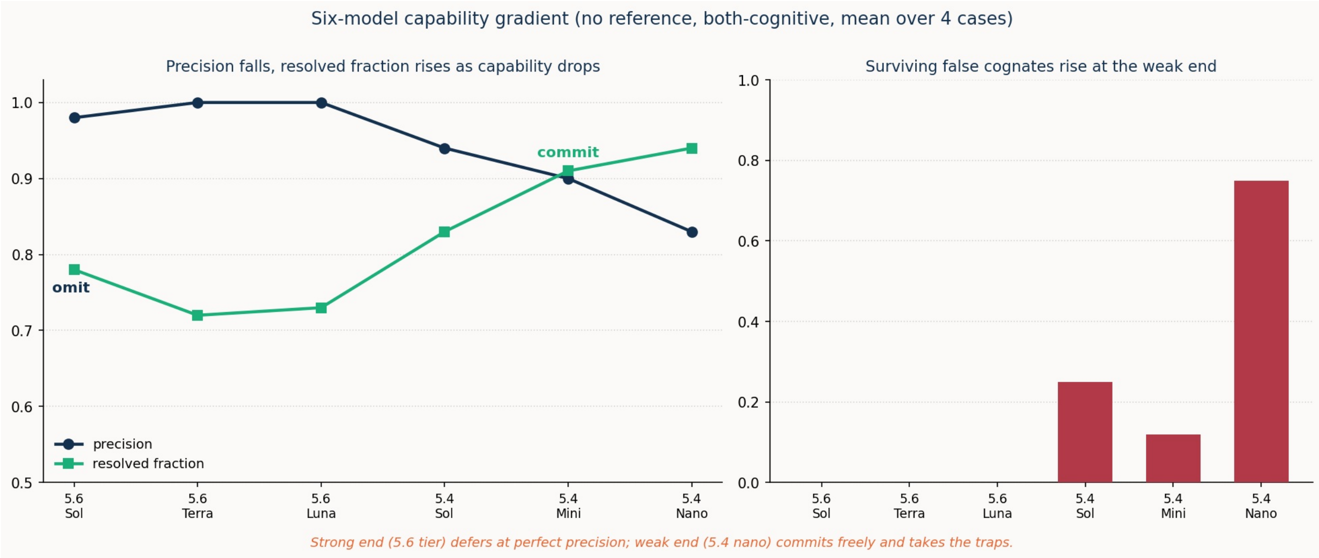

Figure 8. Precision, resolved fraction, and surviving false cognates across a six-model capability ladder (no reference, both-cognitive, mean over the four schema cases). The strong end's omission — high precision, deferral, no traps — shades into the weak end's commission — lower precision, traps taken.

Where a standard *can* pin the distinction, capability decides who can use it — a clean three-rung gradient (Figure 9). On the programme's deepest false cognate, alarm↔anomaly, the strong agent never conflates the two, with or without the reference (intrinsic mastery); the mid agent conflates them once a side is inert, and the RFC 9940-anchored reference **rescues it completely** (the cognate's survival goes 1.00 → 0.00); the weak agent conflates them either way (0.75 with or without — beyond rescue). The lexicon pins the ontology for the middle of the ladder, not the bottom.

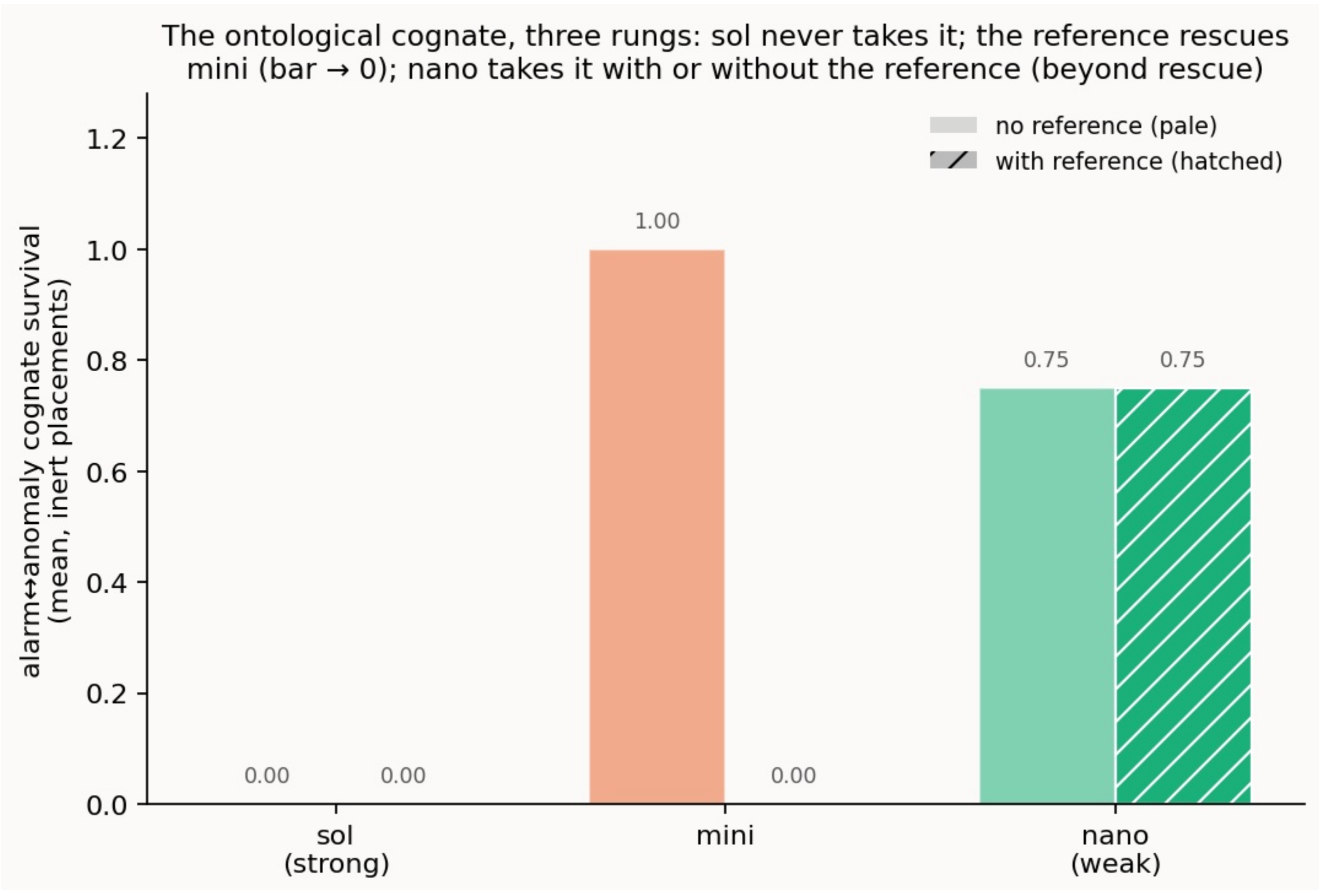

Figure 9 (setting 4). Survival of the alarm↔anomaly cognate at the inert placements, without and with the reference. sol never takes it; the reference drives mini to zero; nano barely moves.

13.3 The reference — what it buys, by component and by placement

Not every thin reference is equal, and the difference is measurable. Ablating the reference's descriptive fields (setting 3, strong agent, opaque identifiers) shows that any *single* field — a shared label, a class, a definition, or an example — is enough to unlock a full close, while a **bare shared identifier with no description is worse than nothing** (precision 0.50, below the no-reference floor): the reference works through shared *description*, never through the pointer. And the same field can *harm* the agent it was meant to help. This shows up in **setting 1**, the configuration reconciliation whose look-alike terms are the programme's richest source of false cognates: there a shallow **class** tag is, for the weak agent, the single worst condition in the programme (Figure 10), driving cognate survival *above* even the no-reference floor — because a class surface reads as evidence for the very cognate it should block.

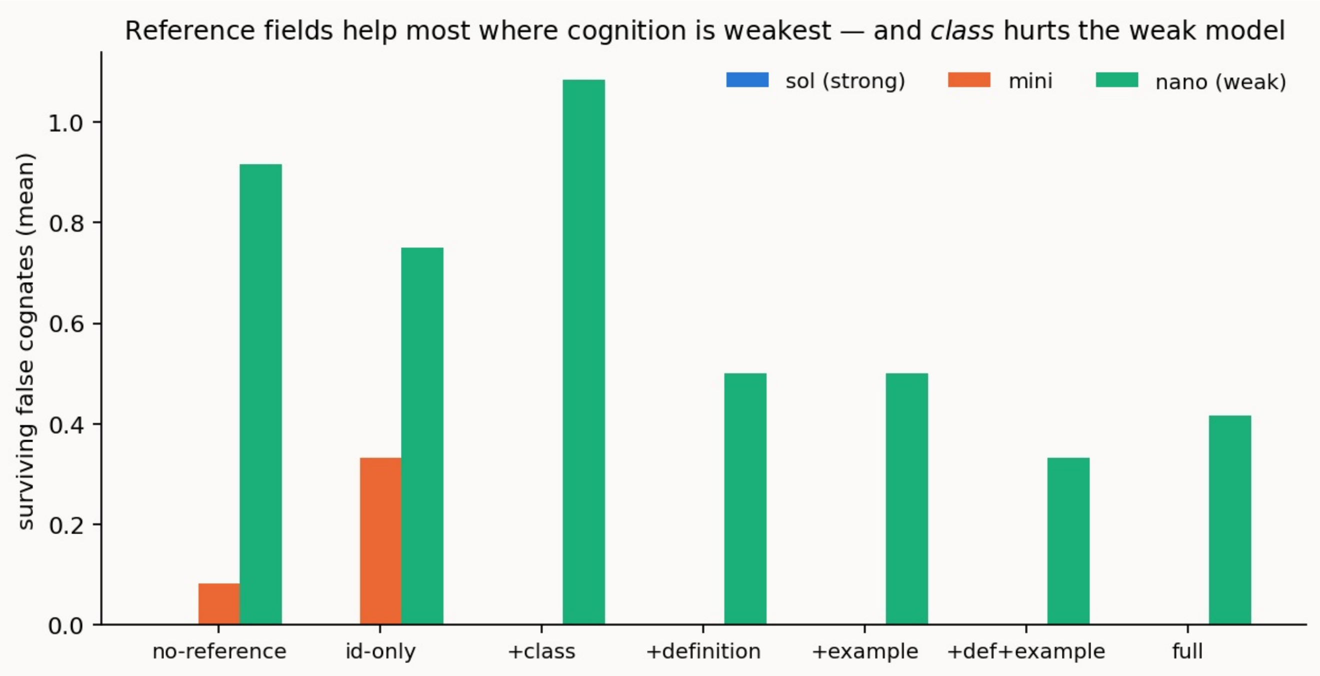

Figure 10 (setting 1). Surviving false cognates by reference content, per model. The strong agent is immune (no bar); for the weak agent the lexical field helps most and the shallow class tag actively hurts — the tallest bar, above the id-only floor.

The reference's two roles run in opposite directions along the spectrum: its **effort** benefit peaks with strong, live cognition (§13.1), while its **correctness** benefit concentrates where cognition — and so verification — is weakest, disciplining the committing agent exactly where it would otherwise err. And independent of any per-reconciliation effect, a shared reference changes how the work **scales** with the number of systems. The unit here is one **reconciliation operation**: a single act of aligning two semantic models, the same operation whose effort §13.1 meters. The question Figure 11 asks is how many such operations it takes to give *N* systems mutual semantic interoperability. Reconciled pairwise, every system must be aligned with every other, which is N(N−1)/2 operations, growing as N^2. Bound instead to one shared reference, each system is reconciled once against the anchor and any two then interoperate *through* it: N operations, growing linearly. This is a structural count, not an experimental average: the number of pairings needed to connect N nodes is a property of the interoperability graph (a full mesh versus a hub-and-spoke), fixed by the topology and exact at every N, which is why the figure is established by construction to N = 12 rather than sampled. Its standing as a measure of *real* work rests on two things. The unit is the very operation the programme meters everywhere else; and the linear branch is licensed by setting 1's own finding that binding to a shared reference is *correct and composes*, so that reconciling A and B each to the anchor genuinely yields an A↔B alignment and the mesh can be collapsed to the hub without loss. Per-operation effort (Figures 5 and 6) is then a roughly constant multiplier, so total cognitive load inherits the N-versus-N^2 split directly: the count is what decides whether the whole workload grows linearly or quadratically.

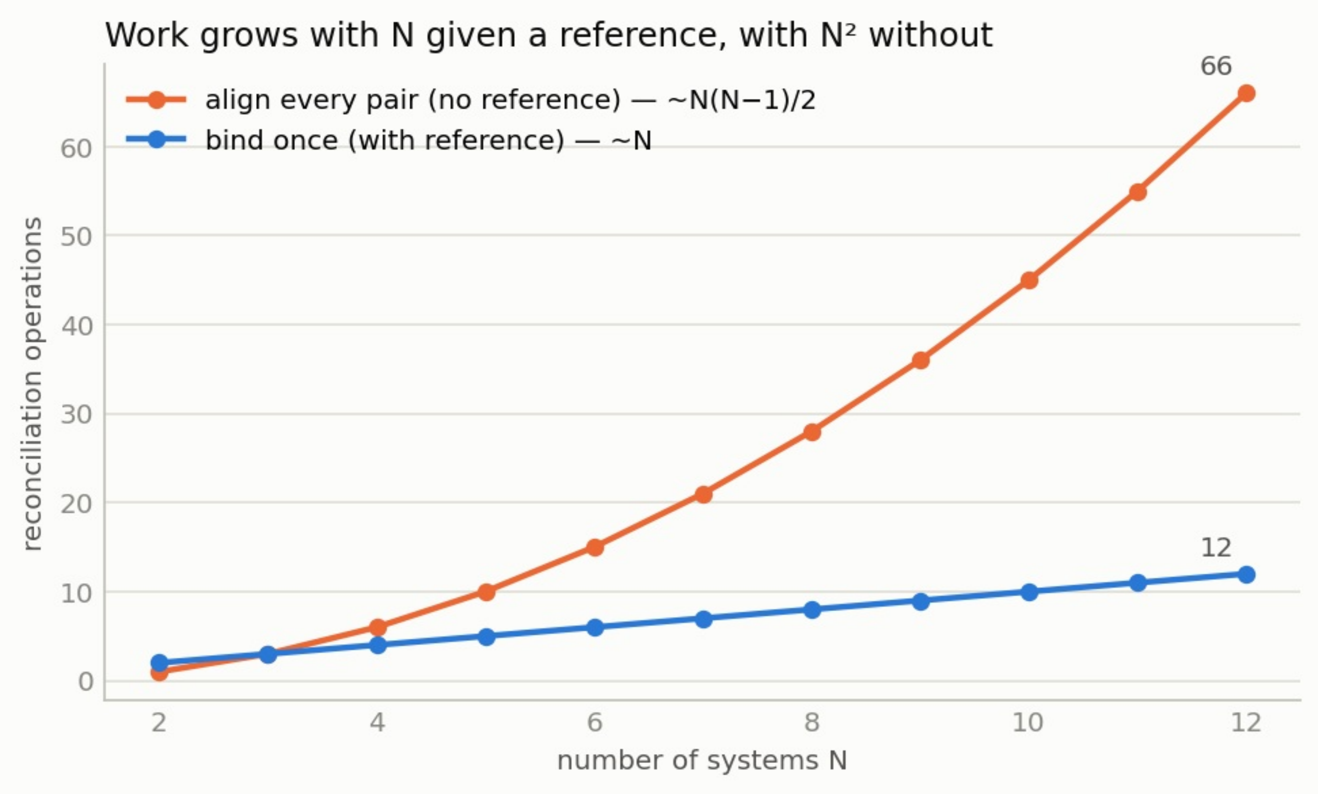

Figure 11 (setting 1, constructed). A count of reconciliation operations (the pairings that must be made) against the number of systems N: bind-once-to-a-shared-reference (~N) versus align-every-pair (~N^2). A structural count established by construction to N = 12, not a measurement of reasoning effort; contrast Figures 5 and 6, which measure cognitive load directly.

13.4 Position on the cognition spectrum — what degrades, and how

Position on the cognition spectrum is the master variable, and moving along it degrades a reconciliation in a specific, measured way. This is sharpest in **setting 2**, the intent setting, where a consumer and a provider negotiate an intent to a workable deal, and the crux is a pragmatic judgement: whether a degraded counter-offer is acceptable to the customer. Across the cognition spectrum (Figure 12) that decision closes autonomously while the customer's judgement is present — live at both-cognitive, or **pre-placed as a portable policy** — and falls to the floor at the mute and both-inert placements, where the correct behaviour is to refer the decision to a person. The pre-placed policy is the mechanism that holds the line where a mute customer cannot: cognition placed in a portable artefact ahead of time closes autonomously what a mute description must hand off.

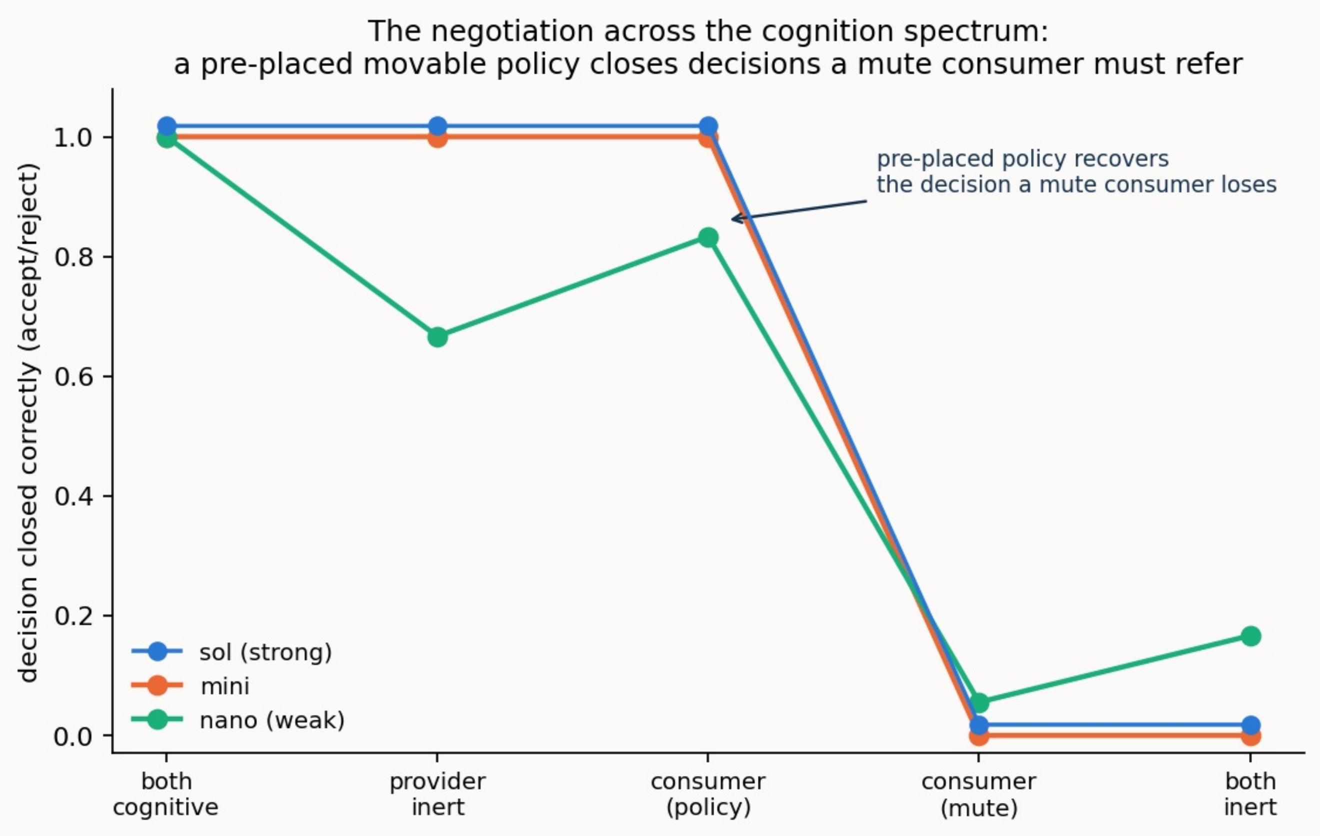

Figure 12 (setting 2). Decision accuracy across the cognition spectrum. It holds high while the customer's judgement is present — live, or pre-placed in a movable policy — and collapses to referral at consumer-mute and both-inert.

The same logic reaches down to the **instance** level in **setting 1**. Instance co-reference is deciding which individual on one side is the same as which on the other. Most pairs are settled from the models alone, but some are genuine look-alikes — structurally identical devices, or same-named but distinct services — that the static descriptions simply cannot separate. Telling those apart means *acting on the live system*: interrogating it, or provisioning something and reading it back. The number of such live probes the agent is permitted is its **probe budget**.

Whether spending that budget actually resolves the hard cases depends on placement, and Figure 13 shows why by crossing the two: it sweeps the probe budget (none, bounded, unbounded) at each of two spectrum placements. At **both-cognitive** the live side is there to be interrogated, so more budget resolves more — the residual falls to zero at unbounded budget. This is what **budget-limited** means: enough probing closes it. At **one-inert** the inert side is a static description with nothing live to interrogate, so the same unbounded budget barely moves the residual (it reaches only 0.08). This is what **structural** means: no budget can close it.

The two are crossed, not confounded — the identical budget sweep is run at both placements — so the curves isolate their *interaction*: probing pays off only where cognition is placed to be interrogated. That is Thesis 2 made concrete on a measured curve. The residual is fixed not by how hard the agent works but by where the cognition sits: at both-cognitive, effort (budget) buys a complete close; at one-inert, no effort can, because the thing that would answer the probe is not there to answer it.

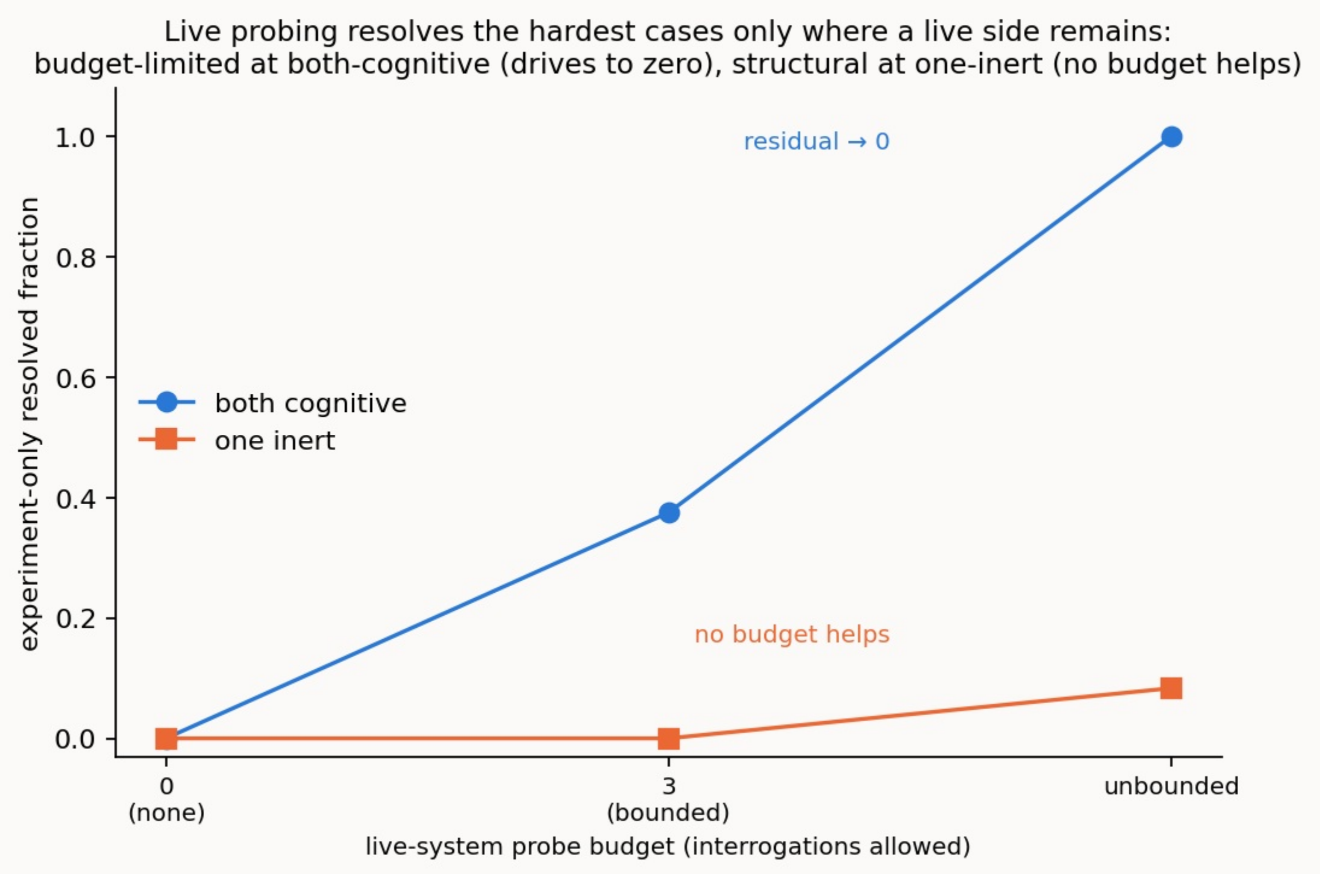

Figure 13 (setting 1). The hardest look-alike individuals, for the strong agent: what fraction gets resolved as the live-system probe budget is swept from none through bounded to unbounded, at two spectrum placements. Budget (the x-axis) and placement (the two lines) are crossed, so the curves isolate their interaction. At both-cognitive a live side can be interrogated and the residual is budget-limited — it drives to zero as probes are spent. At one-inert there is nothing live to probe and it is structural — no budget helps. Same sweep, opposite outcomes, decided by placement, not by effort.

13.5 Reach — how far an agent can see and question

The instance result above raises a question worth separating out. When an agent must *act on the live system* to settle the hardest cases, what limits how far it gets — how *capable* the agent is, or how much it is *allowed to see and to ask*? These are not the same thing. A brilliant investigator handed only a one-line summary cannot do much; a modest one who may interview the witnesses can do a great deal. To tell the two apart, this reading holds the agent at its strongest and holds liveness at the top (both systems reasoning), and varies only what the agent is given to work with, along two axes.

The first axis is **what it can see**: how much of each individual's record is exposed to it — from just an opaque identifier, to that plus a name, plus its attributes, up to the full record including how it connects to its neighbours (its topology). The second axis is **what it can ask** of the live system — from nothing at all (decide from the record as given), to asking about a *named* attribute (where the agent must already know the field name to ask for it), to **discovery** (where it may ask the open question "what facts do you have?" and be told), up to running a small live experiment. As before, the measure is the share of the hardest cases settled correctly — the look-alike pairs that are identical on paper and can be told apart only by questioning the live system, which the study calls the *experiment-only* cases — reported alongside precision, the share of what the agent commits that is right.

The result (Figure 14, left panel) is clean, and because precision stays at 1.00 across the whole grid it is a real gradient, not an artefact of guessing. Seeing more helps, but only up to a point: given no ability to ask, the agent settles none of the hardest cases when it sees only an identifier, and rises to just half of them (0.50) even when it can see the entire static record. Static evidence, however complete, cannot separate twins that differ only in a fact the record does not carry. Being allowed to ask about a *named* attribute barely improves this — the agent has to know in advance which field to ask for. The jump is at **discovery**: the moment the agent may ask the open question "what do you have?", every level of visibility goes straight to a full close (1.00). The decisive reach is not seeing more, and not even asking more, but being able to ask the question you did not know in advance to ask.

The same access lands differently depending on the agent's power (Figure 14, right panel). Handed identical access, the strong agent converts it into a perfect close at perfect precision. The mid agent uses it but plateaus part-way, around 0.67 to 0.75. The weak agent posts high numbers that are not the same achievement: its precision slips (to 0.97–0.98) and it begins committing look-alikes, so its apparent resolution is partly indiscriminate binding rather than genuine settlement — read its resolved fraction without its precision beside it and you would misjudge it, which is the caution the whole study keeps returning to. Power sets the ceiling that reach can reach.

The lesson is as much a design one as a scientific one. How far ad hoc reconciliation carries is not fixed by the agents alone; it is set jointly by how capable they are and by how *interrogable* the thing they meet is — whether the interface on the other side lets a capable agent ask the open questions it needs to. That is a concrete thing a real deployment would have to provide, and it is exactly what the carrier and standards-body engagement discussed under scope could help pin down.

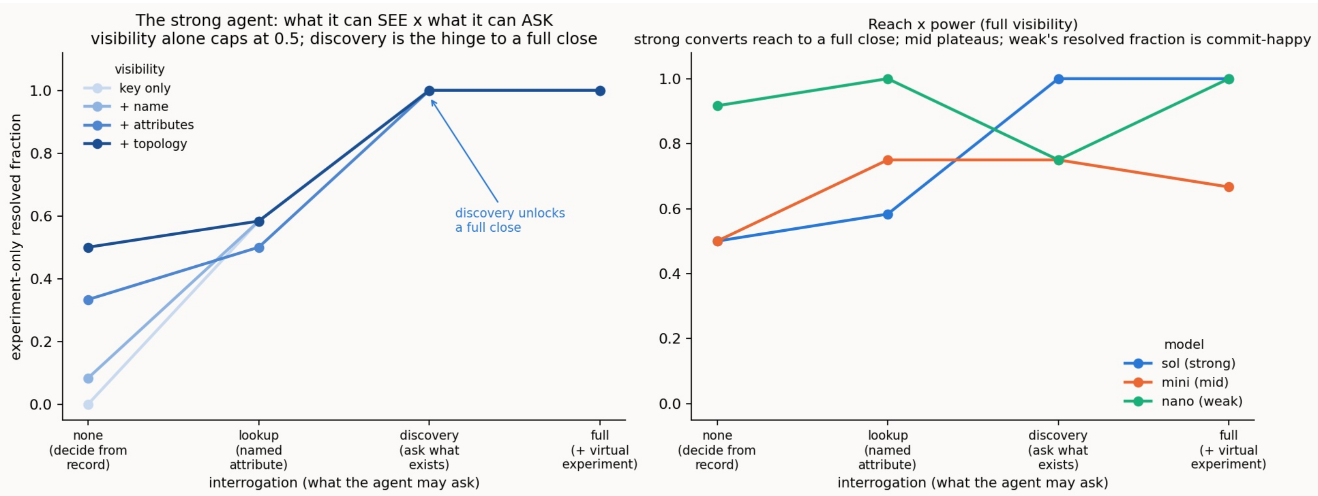

Figure 14 (reach). Left: the strong agent, with liveness held at the top. Each line is one level of visibility (how much of the record it can see); moving left to right widens what it may ask, from nothing, through asking a named attribute, to discovery (asking what exists), to a live experiment. The y-axis is the share of the hardest — experiment-only — cases it settles correctly. Seeing more (higher lines at the left) caps at one-half; discovery collapses that spread to a full close, at precision 1.00 throughout. Right: the same left-to-right axis, now one line per agent at full visibility. The strong agent reaches a perfect close; the mid agent plateaus; the weak agent's high values come with slipping precision (noted in the text), so they are not the same achievement.

13.6 The economics of the reference — when constructing it is worth the cognition

The cross-domain setting (§9) showed that where no standard exists, a capable agent can *build* the thin shared reference itself and then bind through it, lifting a stalled reconciliation to a near-complete close. Building a reference is, however, itself an act of cognition, and that cognition is not free. So a sharper question follows, and it is the one an operator would actually ask: is it worth building a reference every time, or only sometimes? To answer it we ran the same reconciliation three ways on each of the three schema settings, and — this is the new measurement — recorded not just the **close** each reached but the **cognition each spent**, counting the cost of *constructing* the reference separately from the cost of *binding* through it. The three ways are: **no reference** (the agents bind with nothing shared to lean on —

the floor); **constructed** (the agents build the reference themselves, then bind — the standard-free protocol, whose cost we are weighing); and **given reference** (the agents bind through a small, already-published reference — a standard, the cheap upper bound). Two plain reminders, since they carry the result: *resolved fraction* is, of the true correspondences that exist, the share the agent finds and commits — the completeness of the close; and *reasoning tokens* are the model's hidden deliberation, a direct measure of how hard it worked.

The answer is a rule, not a blanket habit, and it has three parts. **Constructing the reference is load-bearing only where no standard exists and the agent is capable.** In the cross-domain setting (two private models, no shared standard) the strong agent with no reference finds only half the true matches — resolved fraction 0.50, because it honestly refuses to guess a seam it cannot be sure of — and building the reference itself carries it to 0.93 for only a little more cognition (about 1,300 reasoning tokens against 1,000 with no reference). There is no cheaper route, because there is no standard to bind through; the construction spend is what buys the close. **Where an effective reference already exists, constructing one is redundant.** In the configuration setting (two public standards) and the observability setting (an RFC-anchored reference), binding through the *given* reference reaches the same close or better for a fraction of the cognition: in configuration the strong agent binds through the given reference to a perfect close for about 166 reasoning tokens, while *constructing* one spends about 1,109 to reach 0.97 — more work, slightly worse result. Building what you already have is wasted effort. **And for a weak agent, construction is actively counterproductive, everywhere.** It is capability-gated: a weak agent builds a poor reference and pays hugely to do it. Averaged over the three settings, constructing the reference costs the strong agent about 960 reasoning tokens, the mid agent about 3,700, and the weak agent about **14,200** — and the close it reaches gets *worse* down that ladder, not better (resolved fraction 0.89, 0.86, 0.72). Handing construction to a weak agent is maximum spend for minimum, or negative, return.

Two things this sharpens. First, it puts a number on how cheap a *given* reference is for a capable agent — tens to low hundreds of reasoning tokens to bind through — which is exactly why a pre-agreed standard is worth so much: someone paid the construction cost once, and everyone binds through it cheaply thereafter. Second, it keeps one honest caveat in view. "Redundant for this one close" is not "worthless": a constructed reference is a durable, reusable, auditable artifact, and a standard is simply a constructed reference that has been *amortised* across everyone who binds through it. So the rule has two regimes — for a one-off reconciliation between capable live agents, build the reference only where no standard exists; but at scale, or over time, building it once and keeping it pays even where any single close did not need it. That, in the end, is the argument for lightweight shared references — constructed ad hoc when none exists, reused when they do — over either a heavy universal standard agreed in advance or paying the construction cost afresh on every exchange.

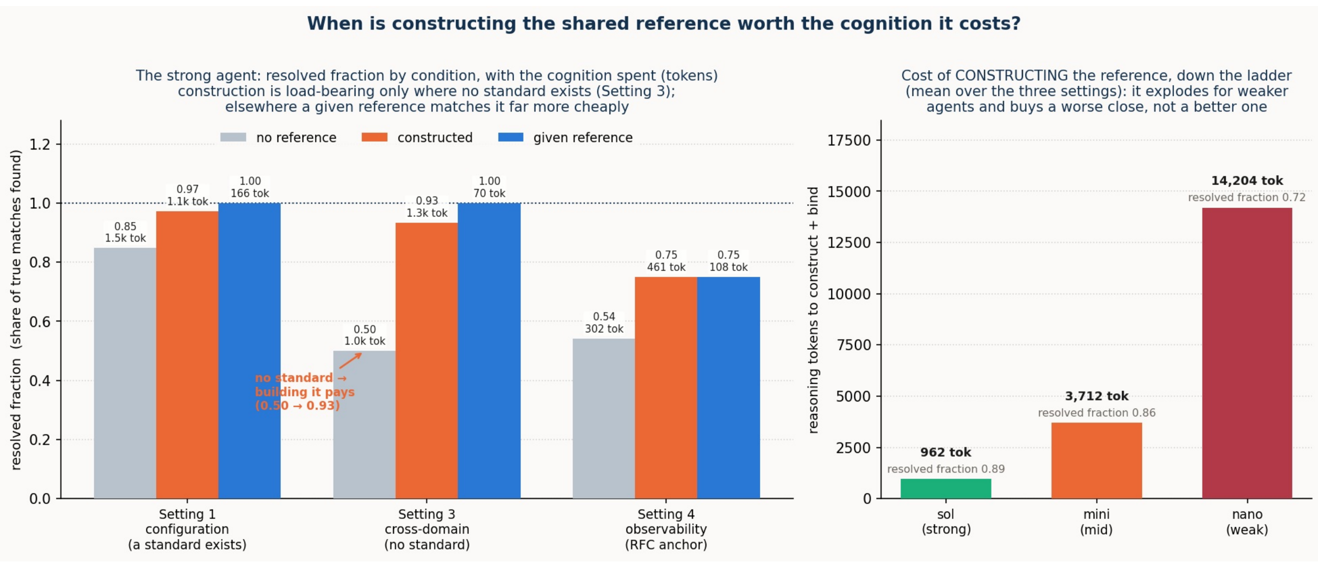

Figure 15 (the economics of constructing a reference). Left: the strong agent's resolved fraction (share of true matches found) under the three conditions — no reference (grey), a reference the agents construct themselves (orange), and a given, already-published reference (blue) — across the three schema settings, with the reasoning tokens each condition spent written on each bar. In Setting 3, where no standard exists, building the reference lifts the strong agent from 0.50 to 0.93 (the arrow) and is the realistic path; in Settings 1 and 4 a given reference reaches the same close or better for far less cognition, so constructing one is redundant. Right: the cost of constructing the reference down the model ladder, averaged over the three settings — it explodes from about 960 reasoning tokens for the strong agent to about 14,200 for the weak one, while the close it reaches (the resolved fraction printed on each bar) gets worse, not better.

14. The maps

The four readings above are the evidence; the two tables below are the index to it. The first locates every finding on the **cognition spectrum** — the master variable — and the second on the **operation** it acts on, so that "cognition matters here" always resolves to a named row.

Table 1 — the cognition spectrum, and what happens at each placement. The rows are the master variable; the columns say what the reconciliation can do, what it leaves in the residual, and what a reference can buy — consolidated across all four settings.

placement	what the reconciliation can do	the residual	what a reference buys
both-cognitive	completes **autonomously** in every setting — bind, negotiate, verdict	none, in principle	partly substitutes for cognition (effort ↓ 1–2 orders); little correctness room left
one-inert	live side reconstructs the mute side; probes gone, so some resolutions turn **structural**	grows — the shortfall appears	supplies missing **information**; rescues the mid agent (ontology, satisfaction)
both-inert	third party can only **propose** for adjudication; no interrogation, no experiment	largest; every judgement referred onward	supplies information, **not authority** — cannot move a judgement
pre-placed policy (intent)	closes the decisions the policy was **authorised** for — pushes the hand-off boundary outward	only the un-authorised judgements	policy carries the **authority**; a reference still cannot

Table 2 — the operations of a reconciliation, and where each was put under test. The rows are the process stages of §3; a cell says how the setting exercised that operation, or "—" where it did not. This is the map for locating any finding on the process: the *here* of "cognition matters here" is a row.

operation (the *where*)	1 · Configuration	2 · Intent	3 · Cross-domain	4 · Observability
Lift	founds it; the lift is the lever (0.66–0.76 → 0.92–0.97)	reused	reused; both sides private	alarm lifted and **decomposed**
Reference construction	given (lexical)	given (unit / invariant)	**constructed** end to end — 0.40 → **0.93**	given (RFC 9940-anchored)
Schema binding (lexical, ontological)	equivalence; reference **substitutes**	shared ground for satisfaction	**the measured stage**: omission vs commission	**ontological** cognate; 3-rung gradient
Attribute pinning	—	bound vs measured metric	committed vs line rate	severity / scores
Instance co-reference	budget-limited → **structural**	measured (endpoints)	bracketed (reproduces s.1)	measured; reproduces s.1
Verification	own step; catches cognates, holds resolved fraction	by **satisfaction**	downstream of the isolated pass	verdict stands in
Pragmatic resolution	left untouched (deferred)	movable **policy**; authority ≠ information	**authority** attribution; reference pins meaning, not authority	**verdict** carries operative meaning; capability-gated
Composition / correlation	—	—	—	robust with the dependency map (all models)
Lifecycle recurrence	—	four-hop loop (self-heal, refer, restore)	—	—
Scaling	linear vs quadratic (N = 12)	—	—	—

15. The surprises

Four results ran against the naive expectation, and they are worth stating as findings rather than smoothing away.

A strong agent scoring *lower* is often a strong agent behaving *better*. Blind at both-inert, the strong agent refuses to affirm what it cannot verify and scores below the mid agent (setting 2's satisfaction 0.29 vs 0.71; setting 3's under-commitment). This is not weakness; it is the honest deferral of Thesis 4, and it is exactly the behaviour a published reference or a live probe converts into a confident, correct close. A raw accuracy number, read without the precision beside it, misjudges it.

A bare shared pointer is worse than no reference at all. One might expect any shared anchor to help; setting 3's ablation shows an opaque identifier with no description dropping precision below the no-reference floor (0.50), because the agent binds by the token and binds wrongly. The reference works through shared *description*, never through the pointer — and the corollary (setting 1) is that a shallow *class* tag, the thinnest description, can mislead the weak agent it was meant to help.

The richer content that raises a weak agent's reach also raises its exposure. The lifted content and instances that let an agent find more correspondences also hand a weak agent more surface to misfire on — its instances turn toxic, over-read as evidence of identity (setting 1's baseline). The lift is unambiguously the lever and unambiguously safe for the strong agent; for the weak agent it is double-edged.

Two operations in the same setting place opposite demands on the agent. In observability, the verdict (a judgement over concern, confidence, and context) is sharply capability-gated, while the correlation (an application of a dependency graph) is robust across the whole ladder. Both resolve significance from the same pragmatic information and both are decisive, but one demands judgement and one is a structural input, and that distinction, not the label "pragmatic," predicts whether a weak agent can do it.

16. Scope, threats, and what remains

The claims here are **existential and mechanistic** — *this is how ad hoc reconciliation works, and here it is working* — not population estimates. Each setting is built around seeded cases, designed to exercise each mechanism and prove each trap rather than sampled from a distribution, with a small number of trials; the reported patterns are the ones stable across the model ladder and the treatment toggles, and the numbers are indicative rather than tight. To guard against any one case being, unknowingly, chosen to work, each setting is exercised on **three independently-built cases**, not one — the breadth reading at the end of this section reports that the findings hold across them. The core ladder is three points spanning a capability range; a six-model sweep resolves the *shape* of the gradient between them, a smooth descent from the strong end's omission to the weak end's commission (§13.2). Golds are derived from the models and validated, which removes drift but leaves the modelling choices — including setting 4's verdict thresholds, stated openly — as authored rather than found. Public standards may have been seen in training, which could flatter the no-reference conditions; the effect relied on throughout is the *difference* a treatment makes under identical inputs.

Two operations bear noting on how they are measured. Setting 3's schema-binding headline brackets the reference-construction step, isolating the worth of that step; the full **construct-then-bind** protocol — the agents building the reference themselves and then closing, with none pre-given — is also run, and it confirms the thesis in the hardest setting: the strong agent lifts from a no-reference resolved fraction of 0.40 to a constructed-reference **0.93**, approaching the reference-given 1.00, at perfect precision and with no false cognate, so constructing the shared ground works and building it is the work (§9). And **instance-level co-reference** is measured in settings 2 and 4 as well as setting 1, reproducing the same budget-limited-then-structural pattern, with a capability gradient in which capable agents resolve fully where a live side can be interrogated while the weakest agent only partially resolves.

The single-case worry, answered as far as in-house work can. The sharpest threat to a mechanistic claim built on one case per setting is that the case was, unknowingly, chosen to work. To test that, two further cases were built for every setting — deliberately different in domain, vocabulary and traps — and the same agents were run on them under the same harness, so that three independent cases now stand behind each setting. The findings reappear on the new cases (Figure 16). In **configuration**, the strong agent again reconciles two new pairs of standard models on its own, and the thin shared reference again mainly serves to prevent the weaker agents' errors. In the **cross-domain** setting, the mirror returns on two new pairs of private, no-standard models: without a shared reference the strong agent under-commits (perfect precision on what it binds, but low resolved fraction because it refuses to guess the seam), and the constructed reference completes the close. In **intent**, working out which offers meet a customer's wish and deciding accept-or-refer under a policy again complete for capable agents, while the multi-hop service lifecycle again grades with capability. In **observability**, the deep alarm-versus-anomaly look-alike is again reliably avoided across two new fault domains. This moves each result from "here it is on one case" to "here it is again on cases built to be different". It does not, and cannot, stand in for real network data: the same hand built the new cases too, so they test robustness to *variation*, not *realism*.

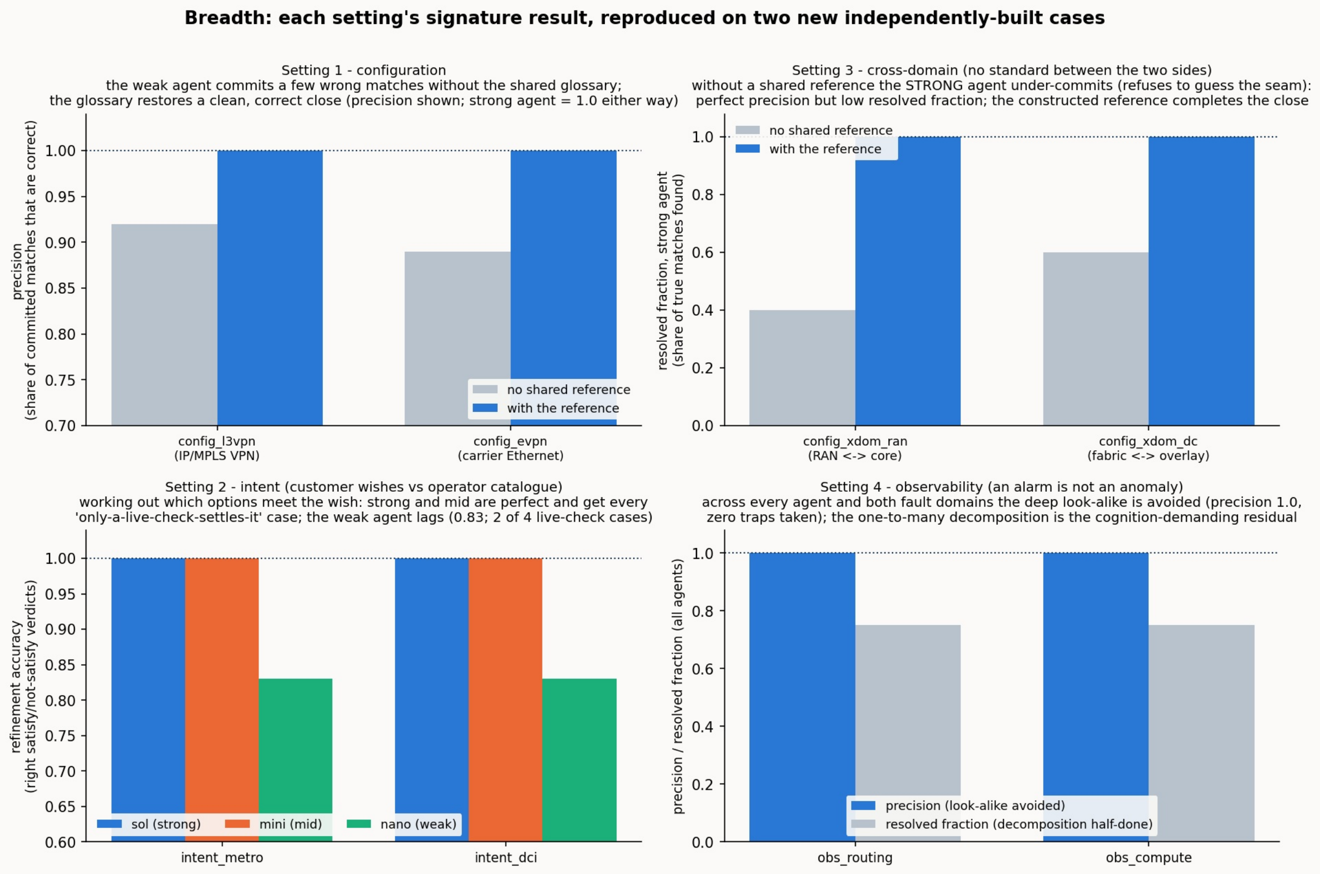

Figure 16 (breadth). One panel per setting. Each shows the setting's signature result on the two new cases built for it, using the real agents scored against the validated answer key. Setting 1: the weaker agent's precision (share of committed matches that are correct) recovers to a clean close once the shared reference is added. Setting 3: the strong agent's resolved fraction (share of true matches found) is low without a reference — it is refusing to guess — and completes with one. Setting 2: the agents' accuracy at working out which offers satisfy the wish, high for the strong and mid agents and lower for the weak one. Setting 4: precision stays at 1.0 (the look-alike is never taken) while resolved fraction sits at 0.75 (the one-to-many decomposition is the residual). The point of the figure is not any single bar but that all four signatures recur on cases built to differ.

What remains is therefore genuinely external — larger and more varied cases drawn from real networks, and the real-data grounding that only carrier and standards-body collaboration can supply.

17. In one paragraph

Two systems that can reason do not need a standard agreed in advance to work together; they can lift their data into portable, self-describing semantic models and reconcile those models ad hoc, for the occasion. Across four settings — configuration, intent, cross-domain, and observability — that is what happens: at the fully-cognitive end the reconciliation completes autonomously in every case, negotiations and significance verdicts included, with no standard and no human. Cognition is what completes it; descriptor methods carry it most of the way, then stop, and the placement of cognition governs how much of the remainder is closed by machine and how much is honestly referred onward. A thin published reference earns its place inside this frame — partly substituting for a capable agent's reasoning, supplying a live agent the facts it can no longer get from an inert side, and, across many systems, making the work grow linearly instead of quadratically — but its reach ends at information: it never provides the authority to decide (whose value governs, whether a trade-off is acceptable), and it must carry meaning rather than a bare pointer to help at all. And the pragmatic layer — what a reconciled thing is for, whether it matters, who decides — is the frontier the descriptor methods never reach: decisive for meaning, carried by cognition or by cognition pre-placed as a policy, and realised only by an agent capable enough to carry it. That pragmatic layer, and the question of how capable an agent must be to work in it, are where the next work lies.